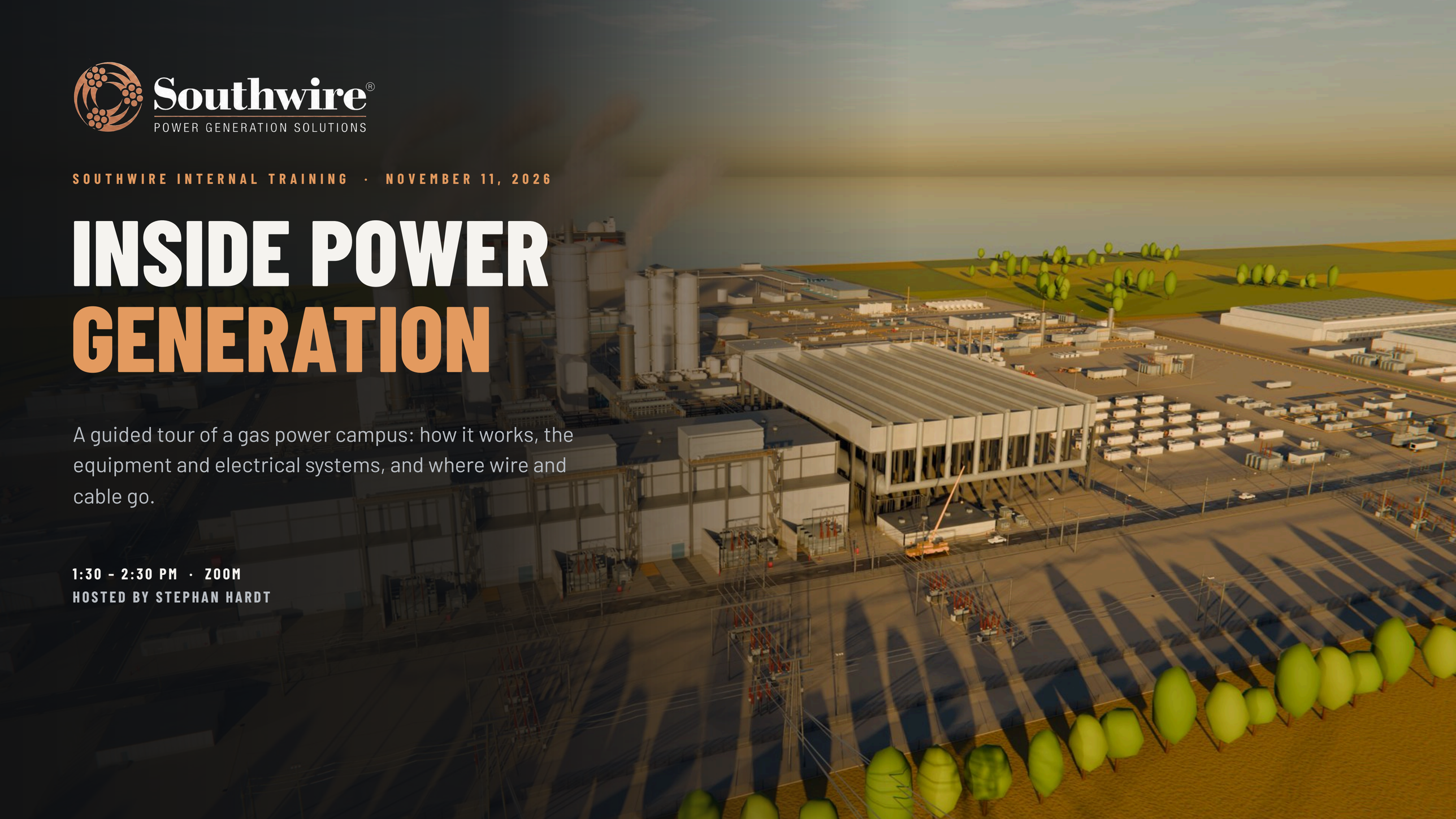
Southwire®
POWER GENERATION SOLUTIONS

SOUTHWIRE INTERNAL TRAINING · NOVEMBER 11, 2026

# INSIDE POWER GENERATION

A guided tour of a gas power campus: how it works, the equipment and electrical systems, and where wire and cable go.

**1:30 - 2:30 PM · ZOOM**
**HOSTED BY STEPHAN HARDT**



TODAY · 60 MINUTES

# THE HOUR

| | | |
|---|---|---|
| 1:30 | **WELCOME**<br>why this matters | 2 MIN |
| 1:32 | **MARKET AND FUNDAMENTALS**<br>how power is made, the gas build; the machines, the grid, cable, who buys | 28 MIN |
| 2:00 | **THE 3D TOUR**<br>seven stops: what it does, its electrical system, its cable | 18 MIN |
| 2:18 | **WIRE AND CABLE TAKEAWAYS**<br>the whole plant in one number, products, pain points | 6 MIN |
| 2:24 | **Q&A**<br>questions from the chat | 6 MIN |

No engineering background needed. At every stop you get three things: **what it does**, its **electrical system**, and **the cable**.

Questions in the chat; we answer them at the end.



stephan.hardt@southwire.com 

# WHAT YOU WILL TAKE FROM THIS HOUR

1

## SALES AND ACCOUNT MANAGERS

Talk plant: name the zones, the buyers and the cable in each. Ask who the EPC is and when the cable package releases.

2

## CUSTOMER SERVICE AND INSIDE SALES

Recognise a power-generation order: MV-105, ARMOR-X, TC-ER, instrumentation by the mile, reels per pull.

3

## SUPPLY CHAIN AND PLANTS

Long runs, many reels, schedule-critical: cut-to-length, SIMpull and on-time delivery decide the job.

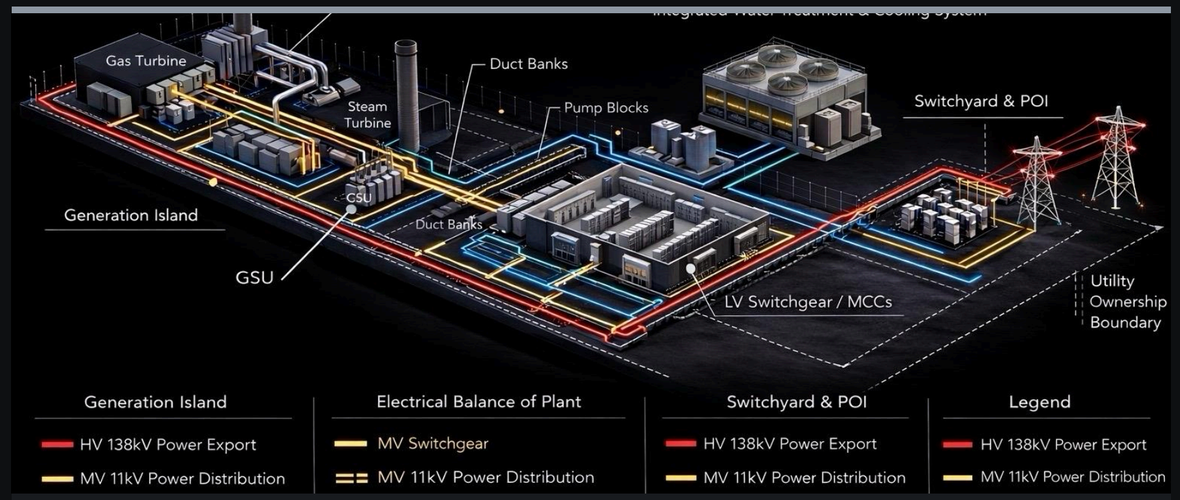
4

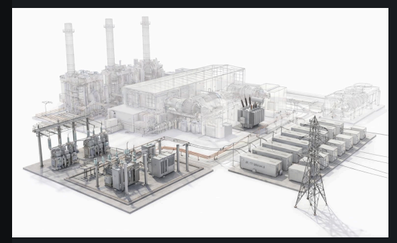
## MARKETING, PRODUCT AND LEADERSHIP

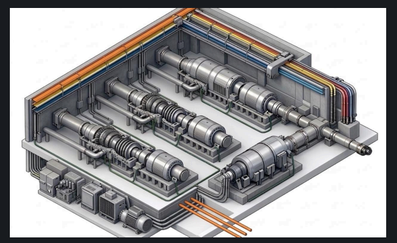
Where the growth is: a boom-scale gas build, behind-the-meter power, and the fleet that needs upgrades.

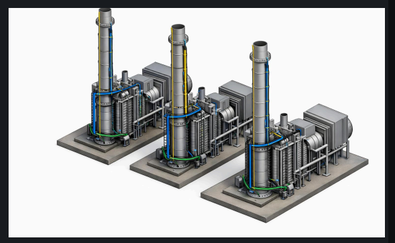
**No engineering background needed.** By the end you can walk a customer through a gas plant, point to where the cable goes, and know who buys it.

# IT STARTED WITH PICTURES THAT LOOKED RIGHT









## V0 FIRST SKETCHES

MARCH

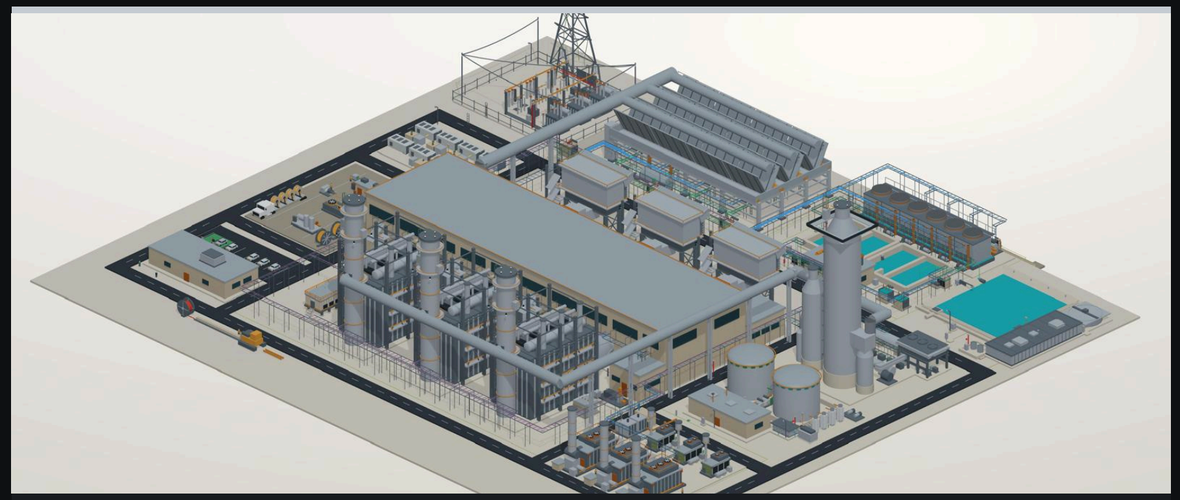
Cable routes colour-coded to tell the story. But generic: non-US voltages (11 kV, 138 kV), invented equipment, no cable counts.

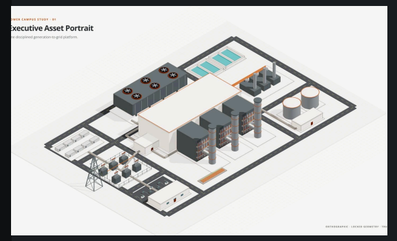
## V1 A BRANDED CONCEPT

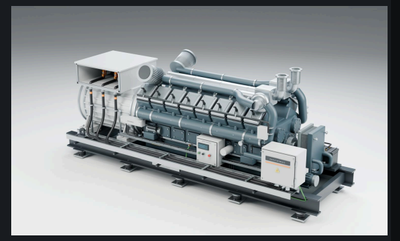
MARCH

A Southwire-branded plant, the turbine hall and HRSGs drawn in detail. Good for a cover, but still invented: nothing you can measure.

# THEN A PLANT YOU CAN MEASURE



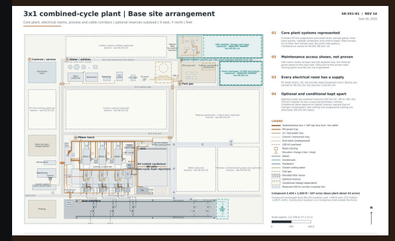


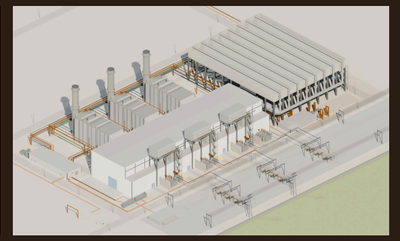


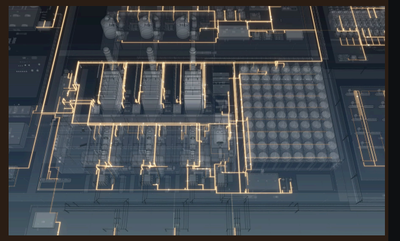
## V2 3D STUDIES

SPRING - SUMMER

Real plant elements in 3D: the hall, gas turbines, HRSGs, engines, the plant zones. Closer, but not to scale and not routed: still no cable to count.





## V3 A PLANT YOU CAN MEASURE

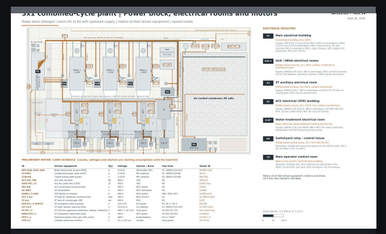
SEPTEMBER - TODAY

Drafted first: the SK-3X1 drawing set (16 sheets, 230 / 13.8 / 4.16 kV, 480 V), then the model built from it. LOD 3, a 6,011-run cable schedule, the X-ray, the tour.

**WHY WE BUILT IT:** SEE THE WHOLE PLANT · MEASURE, DON'T GUESS · ONE SOURCE FOR BROCHURES, DECKS, VIEWER AND SCHEDULE · SPEAK THE BUYER'S LANGUAGE

# CHECKED AGAINST REAL PLANTS, THEN MEASURED

**REAL PLANT DRAWINGS**



## 1 RESEARCH

**700+**

pages of real combined-cycle drawings reviewed: layouts, one-lines, equipment and load lists. They confirmed the equipment, voltages and electrical systems.

## 2 DRAFT

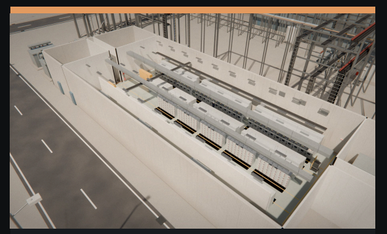
**16**

sheets: the SK-3X1 drawing set, with the power-block electrical layout and a preliminary motor and load schedule.

## 3 BUILD

**1,023**

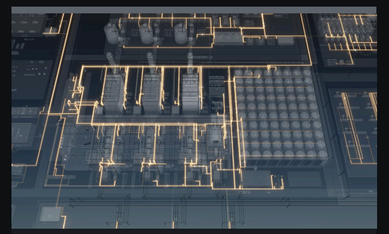
items, 156,247 parts and 719 cable and pipe routes in the 3D model.



## 4 CHECK

**211/211**

footprints match the drawings; 21 heights checked; 1.5 M part pairs tested for clashes: none.



## 5 MEASURE

**21,217**

cables in 6,011 runs, each routed on the model's trays and duct banks: about 1,212 miles.

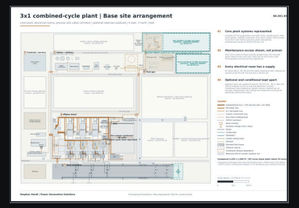
## 6 SHARE

**1**

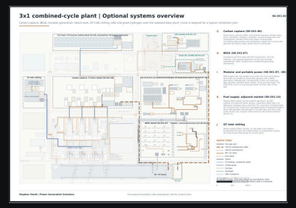
model behind every render, brochure, deck, the 3D viewer, the guided tour and this training.

**Conceptual, not engineered, but not invented.** Every system in the model is one a real combined-cycle plant has, at the voltages and sizes real plants use; the checks run every time the model changes.

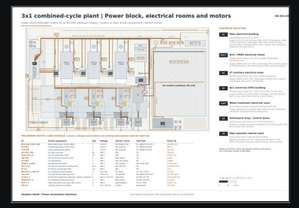
# DRAWN FIRST, NOW AT REV 14



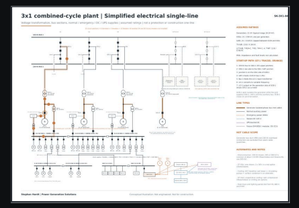

**SK-3X1-01**
Base site arrangement



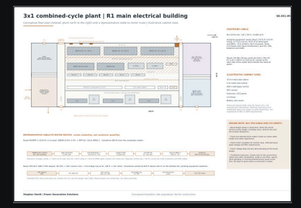

**SK-3X1-02**
Optional systems



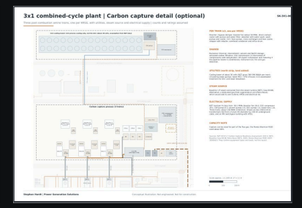

**SK-3X1-03**
Power block, electrical rooms, motors



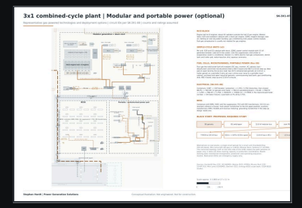

**SK-3X1-04**
Electrical single-line



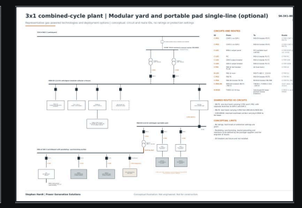

**SK-3X1-05**
R1 main electrical building



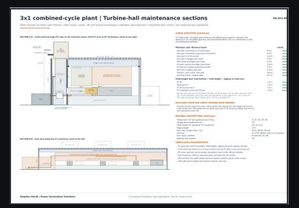

**SK-3X1-06**
Carbon capture



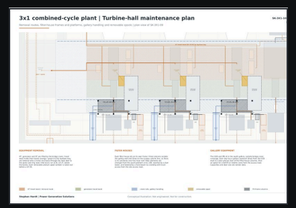

**SK-3X1-07**
Modular and portable power



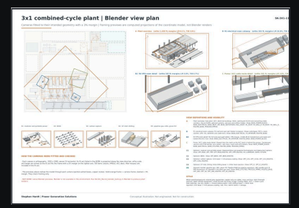

**SK-3X1-08**
Modular yard single-line



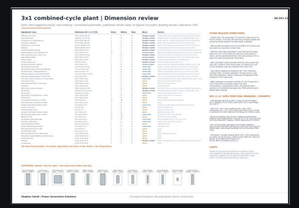

**SK-3X1-09**
Turbine-hall sections



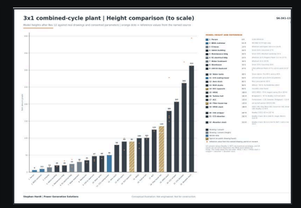

**SK-3X1-10**
Turbine-hall plan



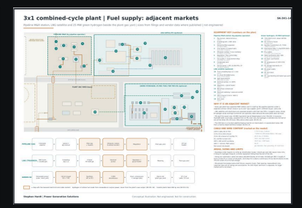

**SK-3X1-11**
3D view plan



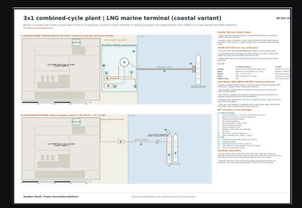

**SK-3X1-12**
Dimension review



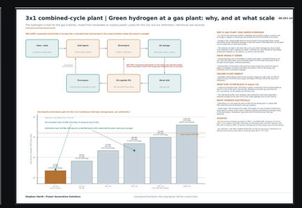

**SK-3X1-13**
Height comparison




**SK-3X1-14**
Fuel supply




**SK-3X1-15**
LNG marine terminal




**SK-3X1-16**
Green hydrogen

## Rev 14

of the SK-3X1 drawing set: 16 sheets, revised as the model found problems. The model is checked back against it on every build.

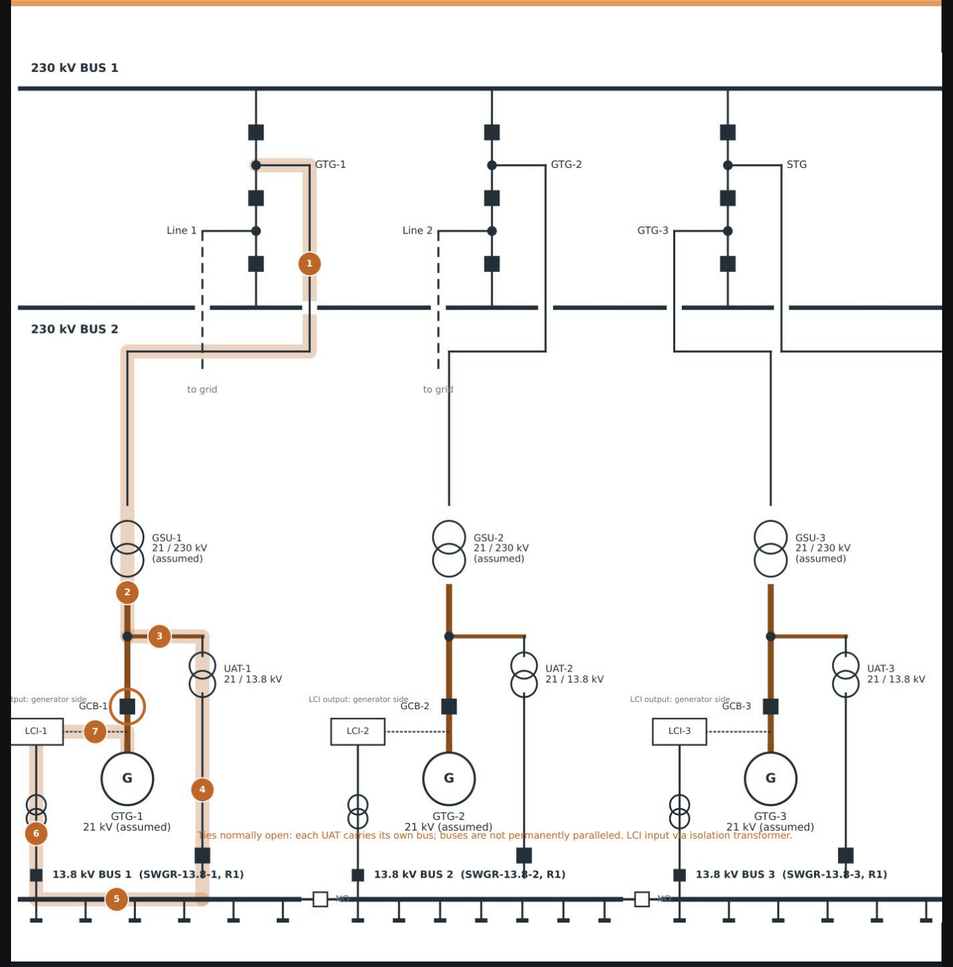
## 700+

pages of real combined-cycle reference drawings reviewed: site layouts, one-lines, equipment and load lists. They set the equipment, voltages and electrical systems.

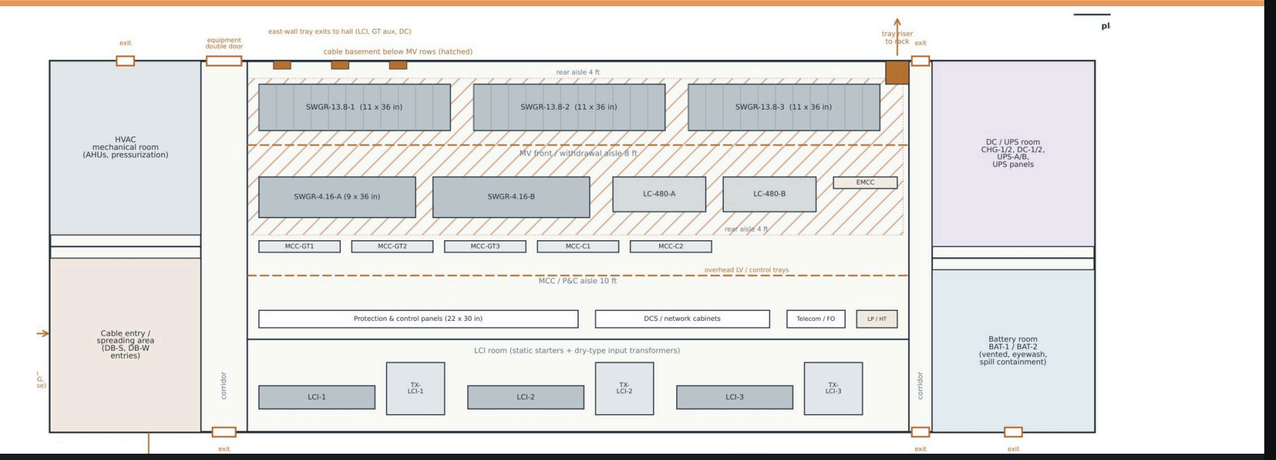
## Tagged

everything beyond Rev 14 is marked as a design change and can be switched off: the modular expansion, the data-centre campus, Boiler Row.

# REAL ENGINEERING DETAIL, SHEET BY SHEET




**SK-3X1-04** Single-line: GT1's start-up path traced, 230 kV down to 13.8 kV



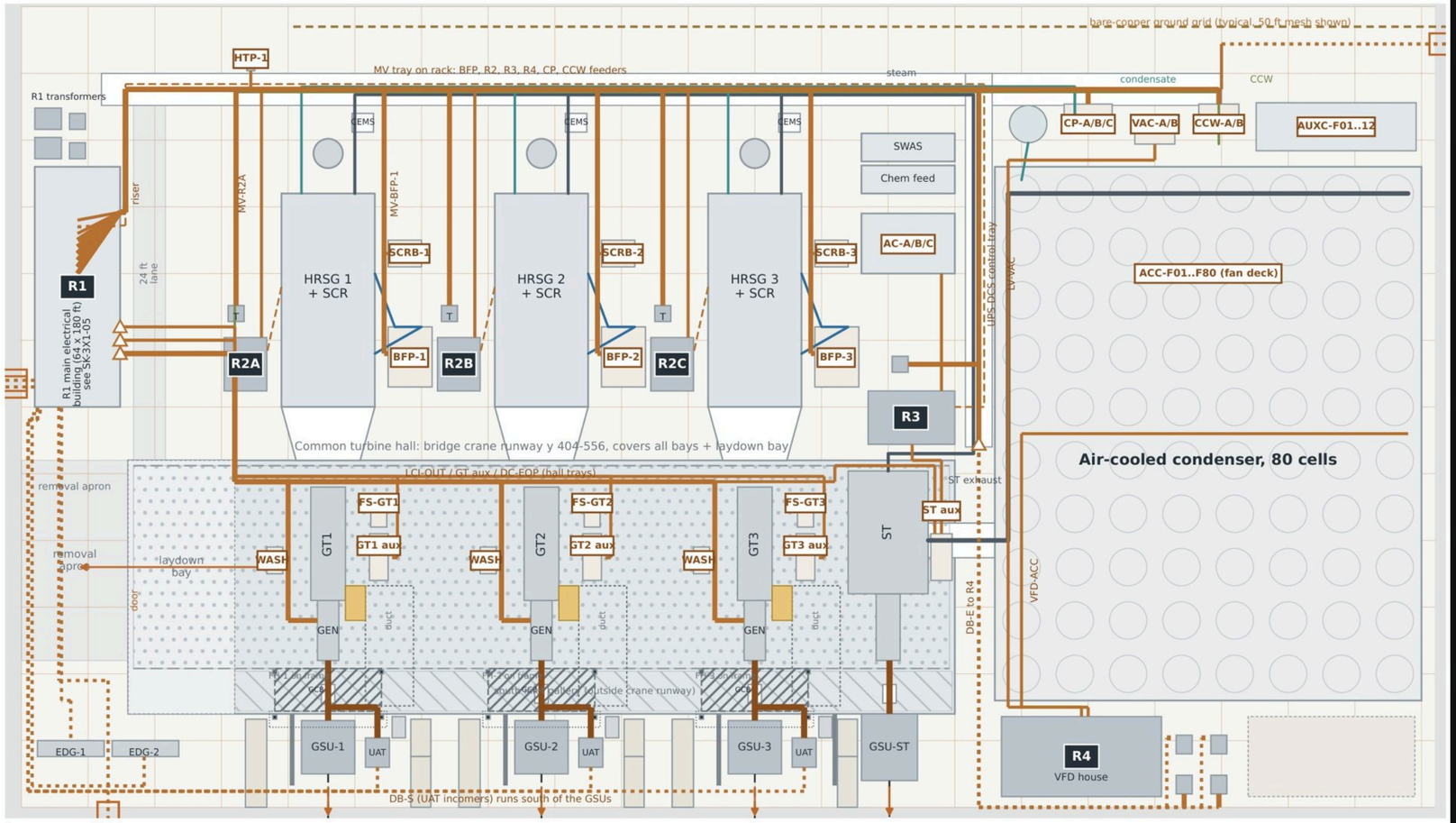

**SK-3X1-05** R1 building: Rev 04 could not hold the switchgear; Rev 05 grew it to 64 x 180 ft

**PRELIMINARY MOTOR / LOAD SCHEDULE (counts, voltages and starters are working assumptions until the load list)**

| ID | Driven equipment | Qty | Voltage | Starter / drive | Fed from | Route ID |
|---|---|---|---|---|---|---|
| **BFP-1A/B, 2A/B, 3A/B** | Boiler feed pumps, at each HRSG | 6 | 13.8 kV | MV breaker, DOL | R1 SWGR-13.8-1/2/3 | MV-BFP-1/2/3 |
| **CP-A/B/C** | Condensate pumps, north of ACC | 3 | 4.16 kV | MV contactor | R1 SWGR-4.16-A/B | MV-CP |
| **CCW-A/B** | Closed cooling water pumps | 2 | 4.16 kV | MV contactor | R1 SWGR-4.16-A/B | MV-CCW |
| **ACC-F01..F80** | ACC fans, fan deck | 80 | 480 V | VFD | R4 | VFD-ACC |
| **AUXC-F01..12** | Aux dry cooler fans (CCW) | 12 | 480 V | VFD | R4 | LV-VAC tray |
| **VAC-A/B** | ACC air-removal vacuum pumps | 2 | 480 V | MCC starter | R4 | LV-VAC |
| **AC-A/B/C** | Air compressors | 3 | 480 V | MCC soft starter | R3 | LV-AIRC |
| **SCRB-1..3 (A/B)** | SCR dilution air blowers | 6 | 480 V | MCC starter | R2A / R2B / R2C | LV-SCR-1/2/3 |
| **GT1-3 aux** | GT lube oil, hydraulics, enclosure fans | sets | 480 V | MCC-GT1/2/3 | R1 | LV-GTAUX-1/2/3 |
| **ST aux** | ST lube oil, turning gear, EHC | set | 480 V | MCC | R3 | LV-ST |
| **EOP-GT1..3, EOP-ST** | DC emergency lube-oil pumps | 4 | 125 V DC | DC starter | R1 DC-1 / DC-2 | DC-EOP |
| **LCI-1/2/3** | GT static starters (start-up drive) | 3 | 13.8 kV in | LCI cabinets | R1 SWGR-13.8-1/2/3 | LCI-OUT-1/2/3 |
| **FS-GT1..3** | GT CO2 fire suppression (detection, release, interlocks) | 3 | 480 V + 24 V DC | local panel | R1 MCC-GT / DC | hall control tray |
| **WASH-GT1..3** | GT compressor water-wash skids | 3 | 480 V | MCC starter | R1 MCC-GT1/2/3 | LV-GTAUX |
| **HTP-1..4** | Heat-trace panels (rack, gas, NH3, tanks) | 4 | 480 V | panel breakers | R1 LC / R-WT | LV-HTP-1..4 |
| **CPR-1/2** | Cathodic protection rectifiers | 2 | AC in / DC out | rectifier | local panels | DC-CP-1/2 |

**SK-3X1-03** Motor and load schedule: every motor, voltage, starter and source

# THE POWER BLOCK, EVERY ROOM AND EVERY ROUTE




**SK-3X1-03** Power block: rooms R1 to R4, every motor at its equipment, every MV and LV route named

## ELECTRICAL FACILITIES

**R1** **Main electrical building**
Conventional building, 64 x 180 ft
Supply: UAT-1/2/3 via duct bank DB-S; EDG-1/2 emergency. Holds 13.8 kV and 4.16 kV switchgear, 480 V load centers, GT and common MCCs, emergency MCC, static starters, UPS, station DC, protection, DCS and comms.

**R2A-C** **Unit / HRSG electrical rooms**
Prefabricated e-houses, 32 x 40 ft, outdoor 13.8/0.48 kV transformer each
Supply: SWGR-13.8-1/2/3. 480 V switchgear, MCC and DCS remote I/O for SCR blowers, ammonia injection, HRSG valves and drains.

**R3** **ST auxiliary electrical room**
Prefabricated e-house, 40 x 65 ft, outdoor transformer
Supply: SWGR-13.8-2. 480 V switchgear and MCC for ST lube oil, turning gear, EHC and air compressors.

**R4** **ACC electrical (VFD) building**
Prefabricated e-house, 60 x 120 ft, four outdoor transformers
Supply: SWGR-13.8-1/2/2/3. 480 V switchgear, 92 VFDs (80 ACC fans, 12 aux cooler fans), MCC for vacuum pumps.

**R-WT** **Water-treatment electrical room**
Room within the water-treatment building (SK-3X1-01)
Supply: SWGR-13.8-2 via DB-W. 480 V MCC for water treatment, wastewater and the fire-pump jockey pump.

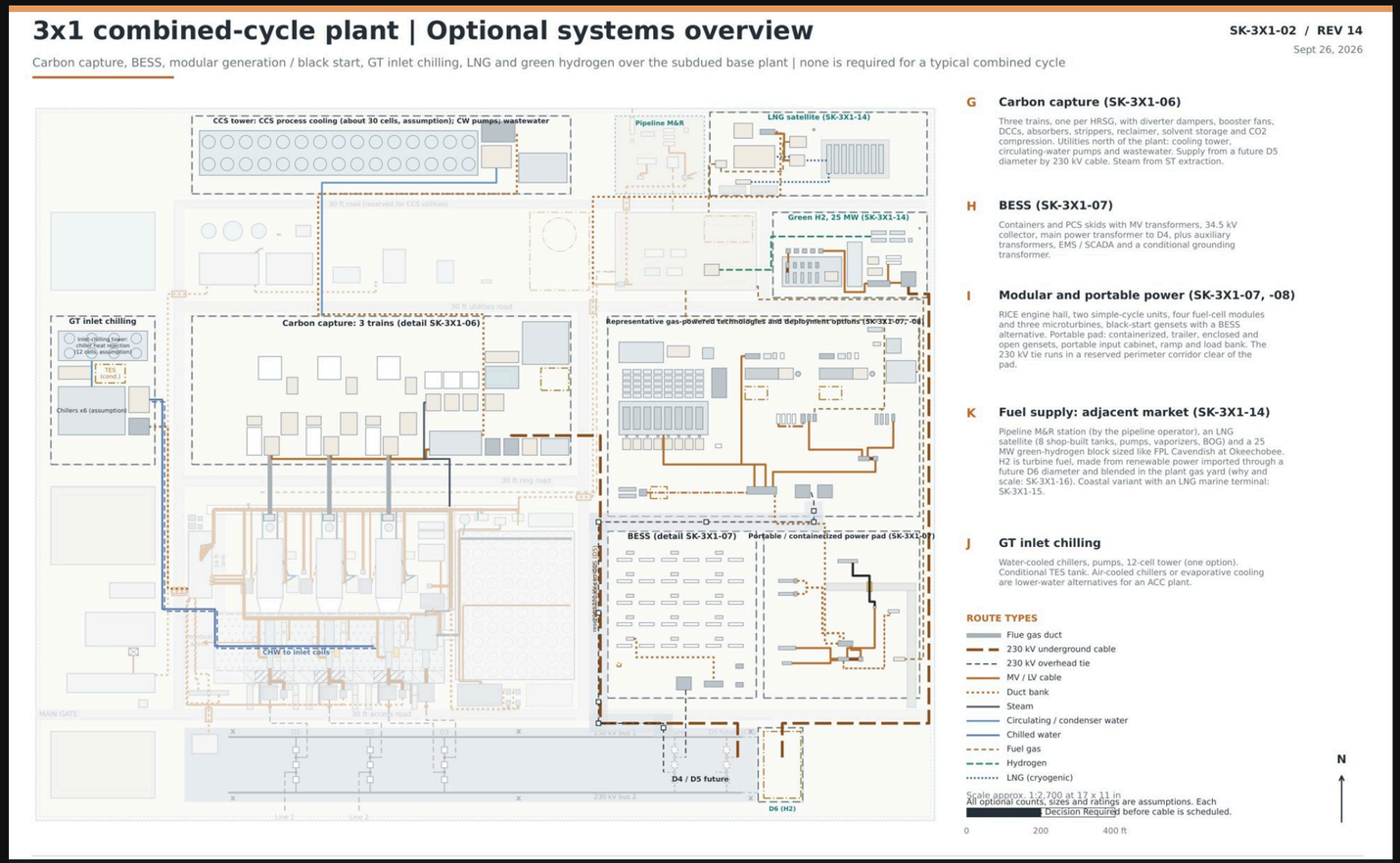
**RH** **Switchyard relay / control house**
Prefabricated control house, 52 x 70 ft (SK-3X1-01)
Protection, SCADA and communications for the 230 kV yard; 125 V DC and fiber to R1 via DB-S.

**CR** **Main operator control room**
Room in the control / administration building
Operator consoles only. DCS cabinets are distributed in R1, R2A-C, R3 and R4. Gas yard: local enclosure, not an e-house.

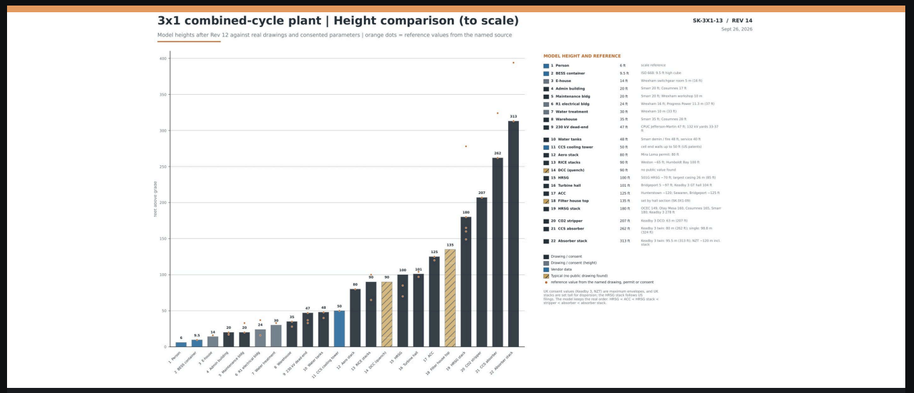
Motors sit at their driven equipment; starters and drives sit in the room named in the table.

Seven electrical rooms: what each one feeds

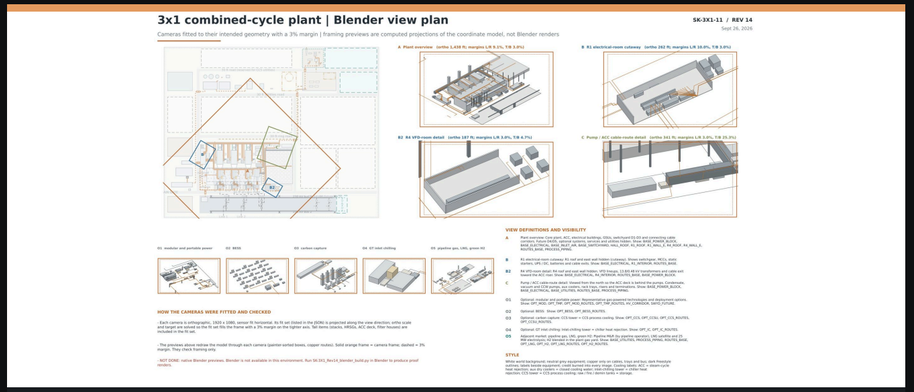
# PLANNED BEFORE IT WAS MODELLED




**SK-3X1-02** Optional systems: CCS, BESS, modular power, inlet chilling, each with its reserved space




**SK-3X1-13** Heights to scale, checked against real references



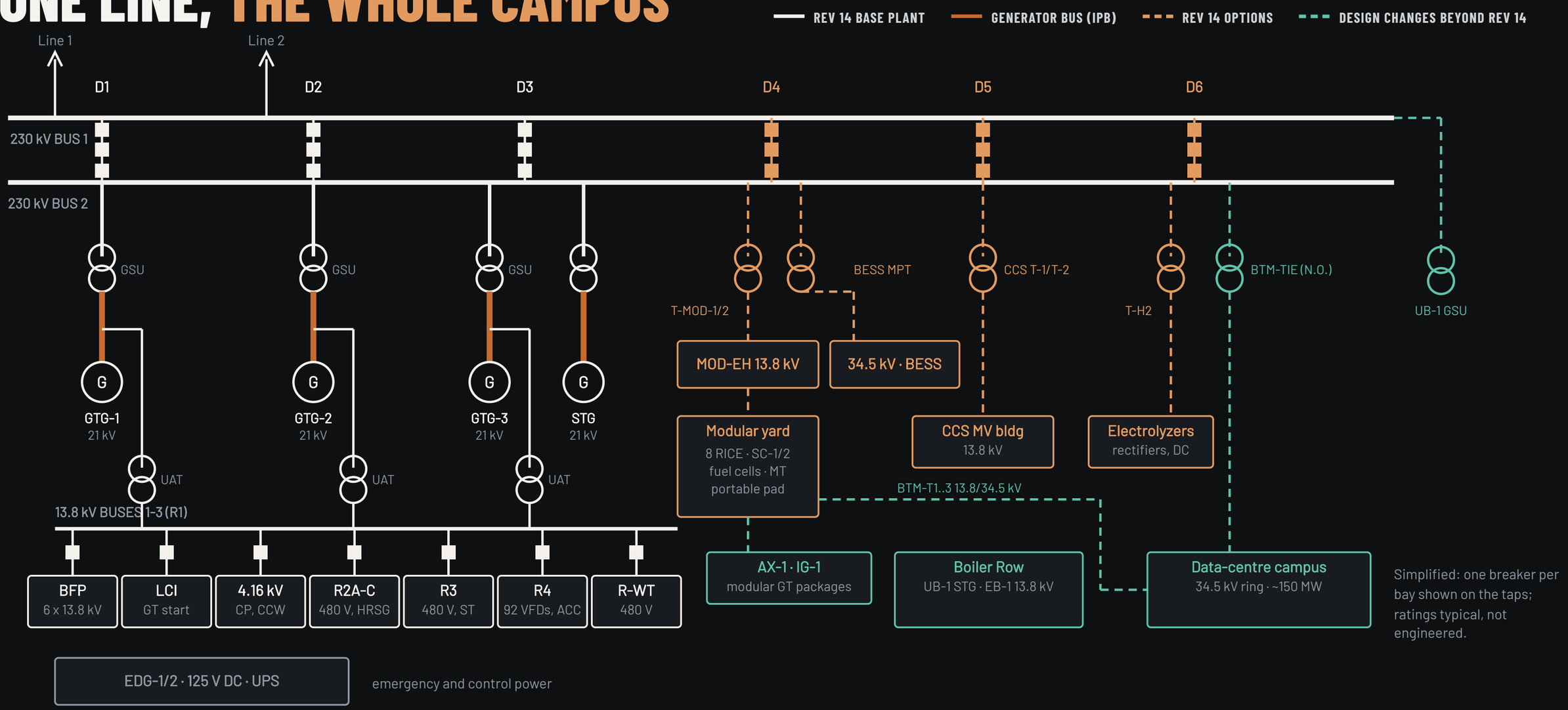

**SK-3X1-11** 3D view plan: every view defined on paper first

**Checked on every build:** the 21 heights on SK-3X1-13 and the 211 footprints on SK-3X1-01 must match the model.



stephan.hardt@southwire.com · INSIDE POWER GENERATION 10/66

# ONE LINE, THE WHOLE CAMPUS



# ITERATED, AUDITED, REWORKED

## SINCE THE FIRST 3D MODEL (29 WORKING DAYS)

**Rev 14**
of the drawing set

**629**
saved iterations of the model

**86**
of them fixes, audits and rework

**1.5 M**
part pairs clash-checked on every build

## FIVE LAYERS OF CHECKS, RUN ON EVERY CHANGE

| | |
|---|---|
| **DRAWINGS** | 211 footprints and 21 heights against the 16 sheets; crane lift clearances in the turbine hall |
| **GEOMETRY** | clashes, floating parts, open pipe ends, anything at grade in a road |
| **ELECTRICAL** | every motor, panel and transformer has a cable; no cable ends in mid-air or runs through equipment |
| **CABLE SCHEDULE** | every run follows a modelled route; sizes from ampacity and voltage drop; totals reconciled |
| **REAL-PLANT LOGIC** | gas safety shut-offs before the burners; utilities buy like utilities; US voltages |

## WHAT THE AUDITS CAUGHT, AND FIXED

- ✓ A gas line routed through two station transformers: rerouted
- ✓ The boiler gas riser upstream of its safety shut-offs: moved
- ✓ Equipment with no cable feeding it, cable ends in mid-air: wired and closed
- ✓ A cooling tower in a different style from its neighbours: rebuilt to match
- ✓ An MV cable tray through the boiler house: rerouted around it
- ✓ Stair flights that never reached the ground: re-seated
- ✓ A duplicate gas-turbine package: removed
- ✓ Cable totals that mixed campus and power block: re-scoped per slide

**DESIGN → BUILD → AUDIT → REVIEW → REWORK, EVERY DAY**

Every number in this training comes from a model that has to pass all five layers before it is saved.



stephan.hardt@southwire.com

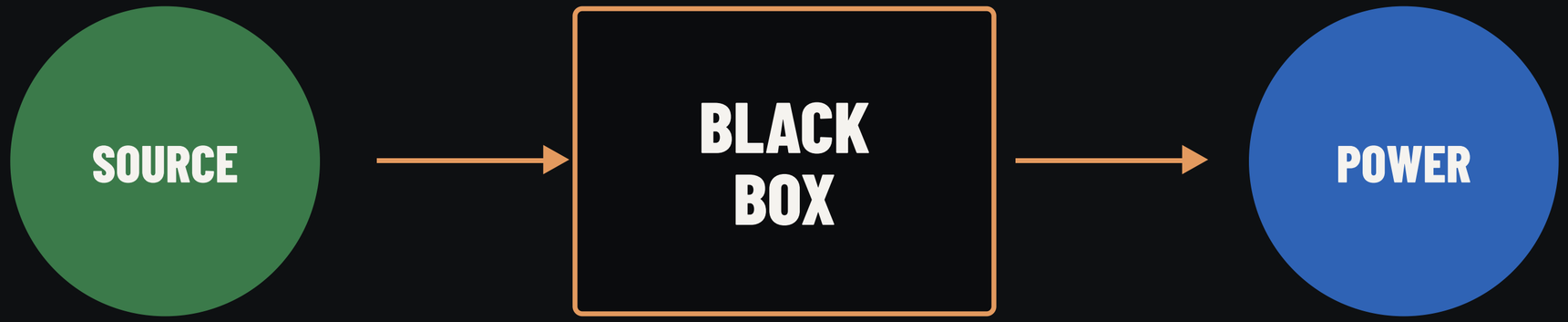
# 01

# HOW POWER IS MADE

Every plant is a black box: what goes in, what comes out, and why we zoom in on gas.

# EVERY POWER PLANT IS THIS

Input → conversion → output. The complexity is what's inside the box, and what's around it.




### INPUT

The energy that goes in: fuel (gas, coal, uranium, biomass) or nature (sun, wind, water).

### CONVERSION

The machine that turns the source into electricity: a turbine and generator, an engine, a solar panel, a fuel cell.

### OUTPUT

Electricity out: to the grid through a switchyard, or straight to a load behind the meter.

Every plant, from a gas turbine to a solar farm, fits this picture. **What changes is the source and the machine in the box.** Next: the four families of power generation, then we open the box of a gas plant.

# HOW ELECTRICITY IS MADE: FOUR FAMILIES

## THERMAL

Heat spins a turbine and a generator.

- **NATURAL GAS** — FIRM
  simple cycle · combined cycle · engines · aero ← today
- **COAL** — FIRM
- **NUCLEAR** — FIRM
- **BIOMASS AND WASTE** — FIRM
- **GEOTHERMAL** — FIRM
- **OIL / DIESEL** — FIRM

## WATER AND WIND

Moving water or air spins the turbine.

- **HYDRO** — FIRM*
- **WIND, ONSHORE AND OFFSHORE** — VARIABLE

Not on our campus; Southwire cables them too: hydro, onshore and offshore wind.

## DIRECT

No turbine: sunlight or chemistry makes DC.

- **SOLAR PV** — VARIABLE
- **FUEL CELLS** — FIRM

SOLAR CARPORT AND EV CHARGERS

In SW-3X1: the fuel-cell bank in the modular yard.

## STORAGE

Shifts power in time; it does not make it.

- **BATTERIES (BESS)** — HOURS
- **PUMPED HYDRO** — HOURS

THE BATTERY YARD

In SW-3X1: the battery (BESS) yard.

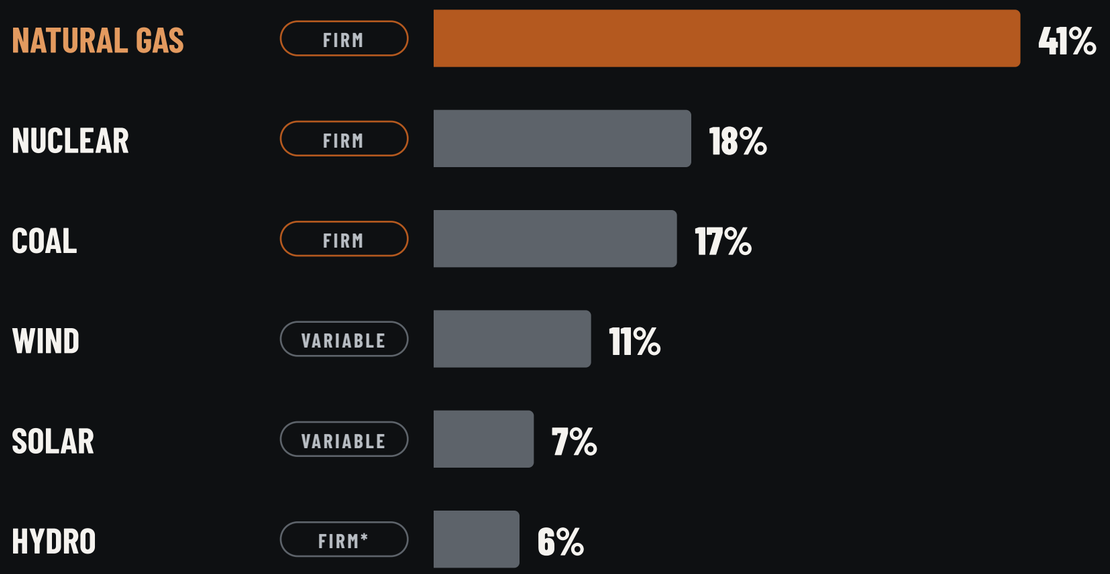
**GAS, ENGINES, AERODERIVATIVES AND PACKAGE BOILERS: ALL THERMAL, ALL ON OUR CAMPUS**

> **Almost all of them spin a magnet inside a coil: a generator.** Solar PV, fuel cells and batteries make DC and need inverters. Firm sources run on demand; variable ones follow the weather. Every one of them ends in cable.

Firm = can run on demand. *Hydro depends on water and season. Storage runs for hours, then recharges.

# THE MIX TODAY, AND WHY WE ZOOM IN ON GAS

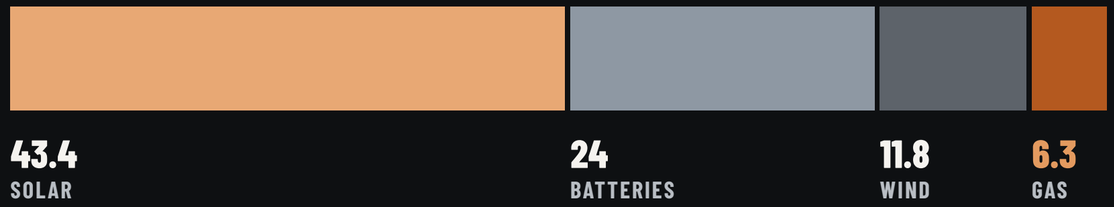
## SHARE OF US ELECTRICITY, 2025




Firm = can run on demand. *Hydro depends on water and season. Batteries store power; they do not make it.

## NEW CAPACITY PLANNED FOR 2026 · 86 GW

Most of what gets added is solar and batteries. Gas is a small share of the GW, and the one that runs on demand.




### FIRM

Gas runs when it is called: day or night, any weather. Wind and solar follow the weather; batteries cover hours, not days.

### THE BACKBONE

41% of US electricity today, and the backup that lets more wind and solar onto the grid.

### THE NEXT WAVE

Data centres need firm power around the clock: 378 GW of gas is in development, much of it built later this decade.

**Southwire cables every one of these:** solar, wind, storage, transmission and distribution. **Today we focus on gas**: firm power, a boom-scale build, and some of the most cable-dense plants on the grid.

EIA, Electricity explained (2025 utility-scale generation, rounded) and Today in Energy, Feb 2026 (planned 2026 additions; developers' plans, not all will be built).

# 02

# THE MARKET

A boom-scale gas build, and cable is on the critical path.

# A BOOM-SCALE GAS BUILD, AND CABLE IS ON THE CRITICAL PATH

THE MARKET

**378 GW**

US gas power in development; 52 GW already under construction

THE CABLE

**~$1.2B**

core wire & cable, 2026–2035, at $8–13M per GW

THE PIPELINE

**386**

gas projects we track in 40 states, every owner type

THE TRACTION

**$50M+**

booked in Q3; $30M in September alone

## TODAY'S GOAL: KNOW THE PLANT, SO YOU CAN FOLLOW THE CABLE THROUGH IT

Global Energy Monitor, Aug 2026; Southwire PowerGen planning model and internal figures, 2026.

MARKET CONTEXT

# LOAD GROWTH IS PULLING GAS POWER FORWARD

**166 GW**

US peak load growth forecast by 2030

about 90 GW of it from data centres

**378 GW**

US gas power in development, mid-2026

189 GW tied to data centres

**52 GW**

under construction, up 76% in H1 2026

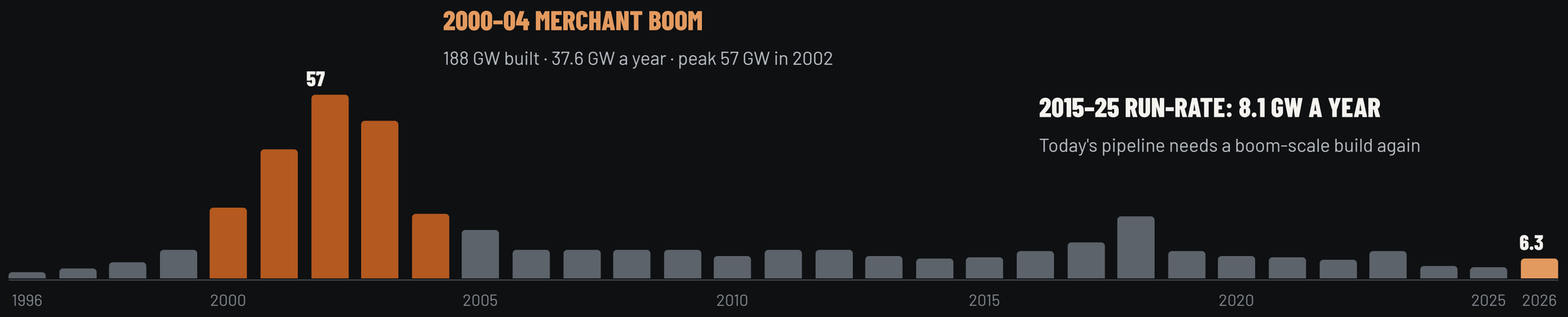
the largest gas build in the world

**Up to 7 yrs**

quoted lead time for some large gas turbines

procurement decisions are being made now

US GAS CAPACITY ADDED PER YEAR, GW · PROVISIONAL




Grid Strategies, 2025; Global Energy Monitor, Aug 2026; S&P Global, May 2025. History: EIA and Southwire provisional series, to be verified.



stephan.hardt@southwire.com 

OEM ORDER BOOKS

# TURBINE BACKLOGS ARE FULL

The three large-frame turbine OEMs together hold **about 220 GW** of gas turbines in backlog and slot reservations. Modular and packaged gas power is growing beside them.

MODULAR

**+72%**

power-generation sales at the largest engine-generator maker, Q2 2026

BEHIND THE METER

**2.3 GW**

modular gas fleet for Texas AI data centres, plus 1 GW for another campus

PACKAGED

**500 MW**

standard modular gas-turbine plant, offered as a product

OEM releases, Q2–Q3 2026 (firm backlog plus slot reservations; gas-turbine backlog; large-frame backlog); Utility Dive, Aug 2026; POWER, Oct 2025 and Feb 2026.

SIZING THE OPPORTUNITY · CONSERVATIVE PLANNING CASE

# ~$1.2B OF CORE GAS CABLE, 2026–2035

**CAPACITY**

## 93.7 GW

modeled additions through 2036, from the 378 GW pipeline: 90% of the 52 GW building, 13.5% of the rest

**PLANT INVESTMENT**

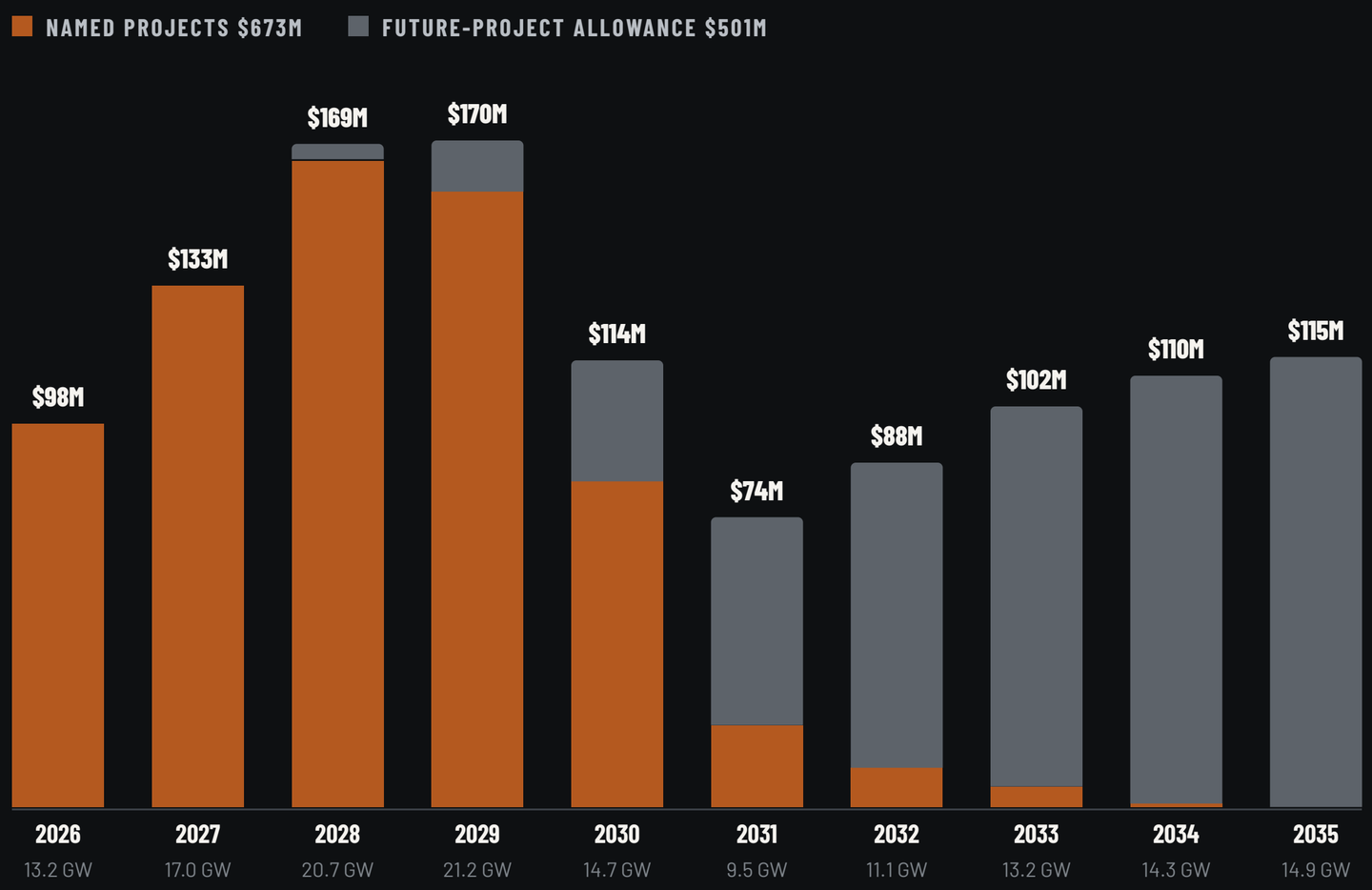
## ~$180B

93.7 GW × $1,923/kW blended (combined cycle, peakers, engines)

**WIRE & CABLE**

## $0.75–1.2B

$8–13M per GW: under 1% of plant capex, on the critical path to first power

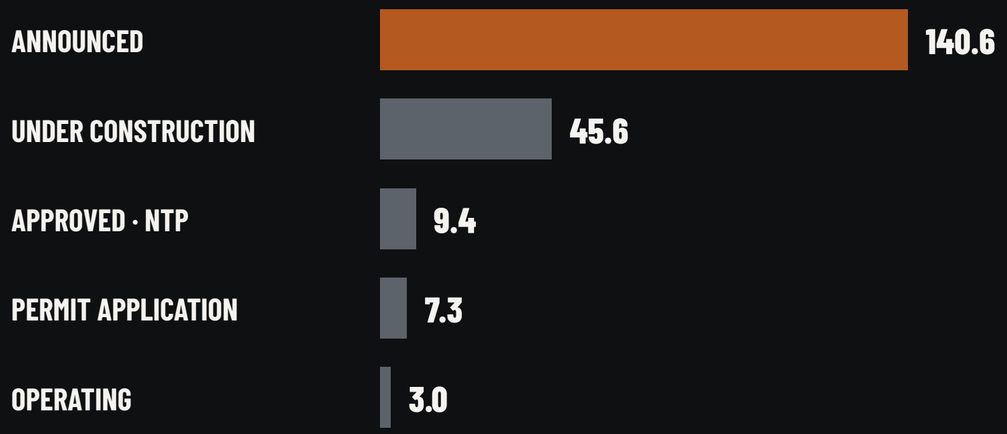



Core cable by purchase-order year; below each year, modeled PO-year GW (not installed GW). GW × $/kW × addressable share × win probability. OEM internal wiring and HV excluded.

Global Energy Monitor, Aug 2026; completion rates, blended cost and cable rates are Southwire planning assumptions.

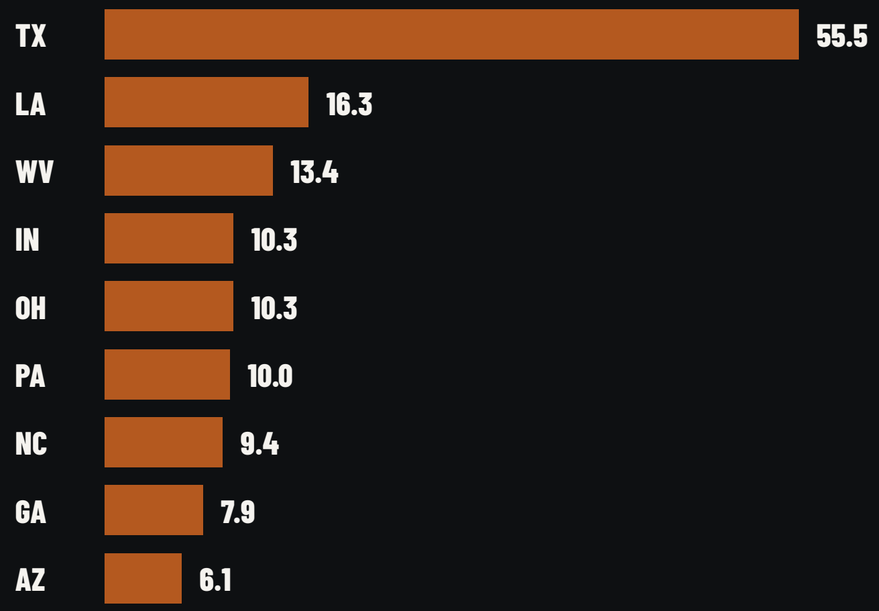
# 205.8 GW ELIGIBLE, MOST OF IT STILL EARLY

BY STATUS · GROSS GW, COD 2026-2035




**68%** **of eligible capacity is only announced.** Being on the register does not mean a cable package is open; it means a conversation to start.

TOP STATES · GROSS GW, 386 PROJECTS



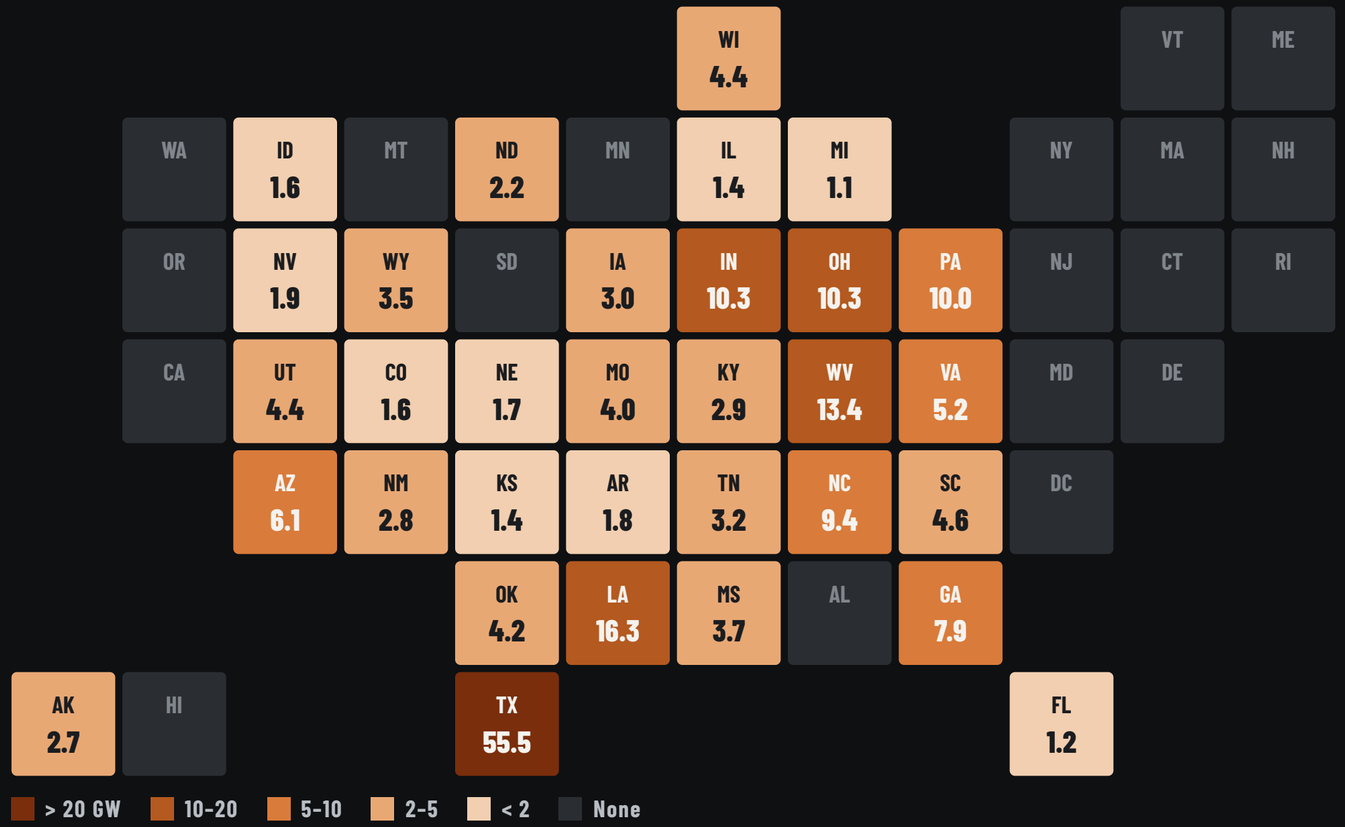

Greenfield 137 GW, brownfield 69 GW. Utilities reuse old sites (coal-site repowers, expansions); independent and behind-the-meter developers mostly build new, often on data-centre campuses.

Southwire PowerGen register, Sep 11, 2026: 386 records across 305 site groups; gross GW, unweighted.

PROJECT CONCENTRATION

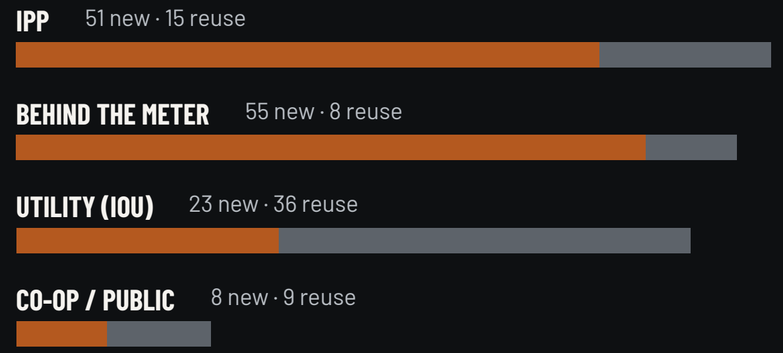
# WHERE THE BUILD IS, WHO OWNS IT, WHAT KIND OF SITE

## GROSS GW BY STATE · 386 PROJECTS, COD 2026-2035




## OWNER × SITE · GROSS GW

GREENFIELD 137 · BROWNFIELD 69




### UTILITIES REUSE OLD SITES; IPPS AND BTM BUILD NEW

| | |
|---|---|
| Greenfield, grid-connected | 77.0 |
| Data-centre / BTM campus | 54.5 |
| Coal-site repower | 41.3 |
| Expansion at an operating plant | 21.4 |
| Industrial captive and site reuse | 11.6 |

Southwire PowerGen register, Sep 11, 2026. Gross GW, unweighted. Site classes provisional (Rev01).

# 03

# INSIDE A GAS PLANT

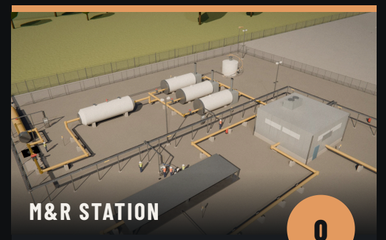
The machines, the grid, and the cable that ties them together.

# OPEN THE BOX: THE PHYSICAL ANATOMY

**INPUT**
fuel / energy source

**BLACK BOX**
thermal conversion

**OUTPUT**
dispatchable power, delivered




## FUEL INFRASTRUCTURE

- Gas production, pipelines, compression
- Storage and LNG logistics
- Plant-gate fuel delivery

## FUEL INPUT

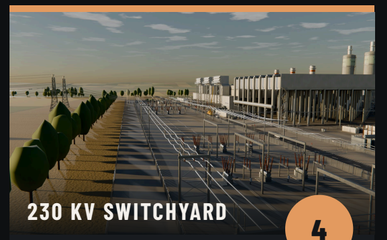
- Natural gas, coal, oil / diesel
- Hydrogen blends
- Biomass where it applies

## THERMAL CONVERSION

- Frame and aero gas turbines, engines (RICE)
- Combined cycle: GT + HRSG + ST
- Modular and H2-ready packages

## ELECTRICITY OUTPUT

- Dispatchable AC power
- Stable, reliable load
- Baseload, mid-merit or peaking




## DELIVERY & FLEXIBILITY

- Transmission and distribution
- MV / LV delivery, behind-the-meter on site
- Storage and flexibility support

## END USE / OFF-TAKE

- Utilities, IPPs, industrial loads
- Data centres and other off-takers
- Critical infrastructure

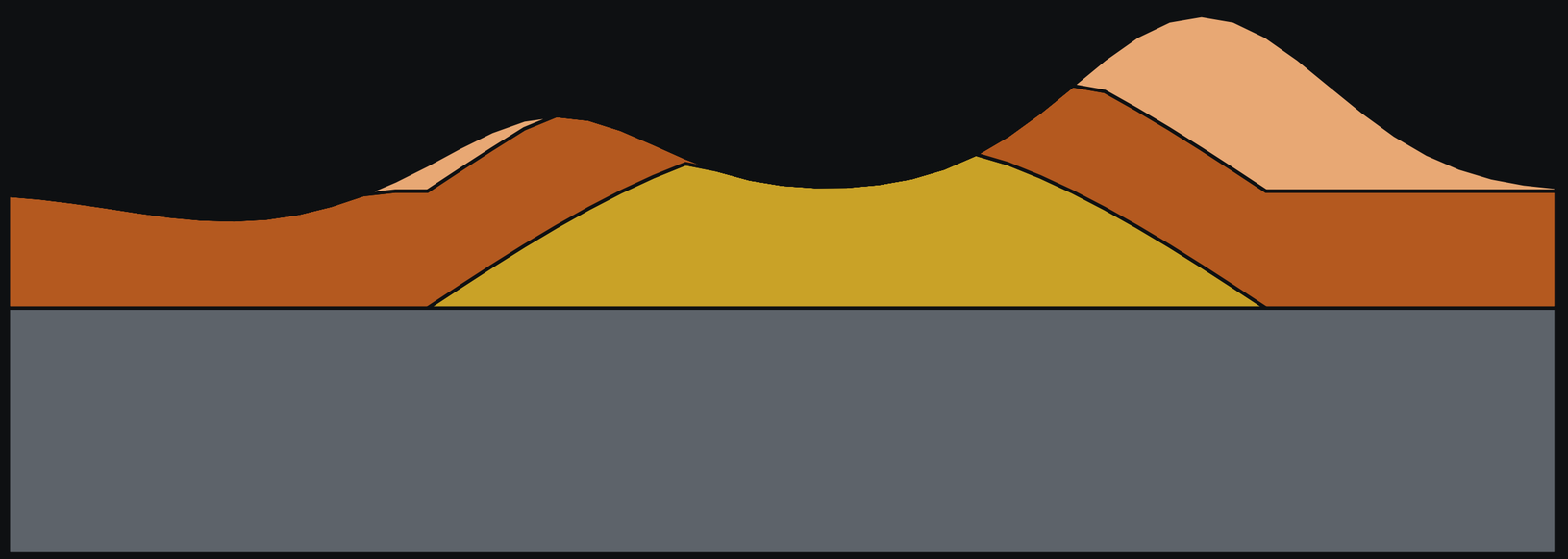
Back to the black box, now for gas. Gas turbine, combined cycle or engines: **the same physics on every project.** What changes is the machine in the box, and who owns it.



stephan.hardt@southwire.com 

# A DAY ON THE GRID: WHO RUNS WHEN

Demand rises in the morning and peaks in the evening. Each kind of plant takes a layer.



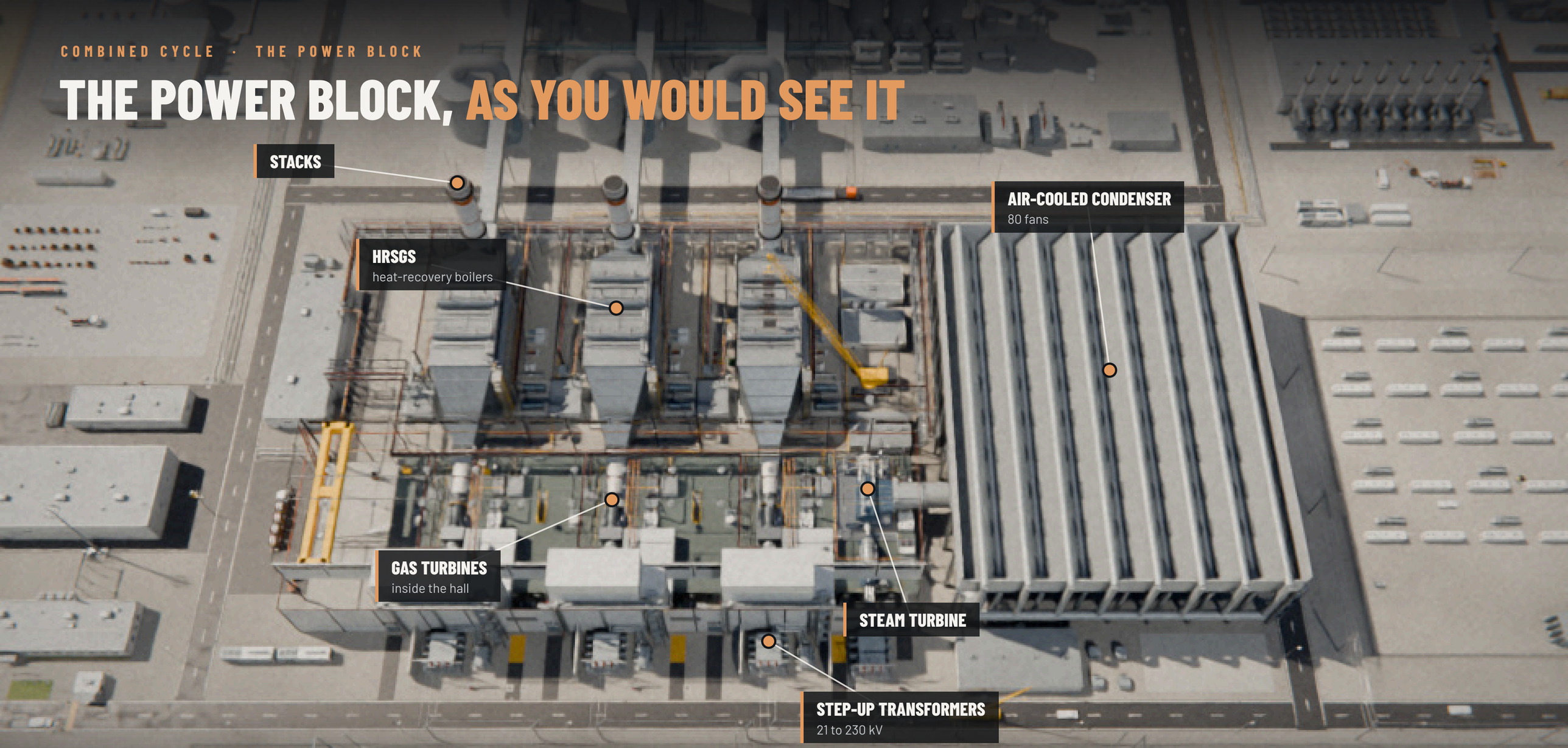

- Peaking: simple cycle, engines, batteries
- Mid-merit: combined cycle
- Solar
- Baseload: nuclear, coal, combined cycle

The evening peak arrives as the sun sets: that is when gas and batteries earn their keep.

Illustrative summer weekday, not a real grid's data.

**Every start and stop cycles the plant**, and its cable: heat, vibration and terminations that move. Peakers may start every day.

# THE POWER BLOCK, AS YOU WOULD SEE IT




Steel, concrete and machines: the part everyone photographs. **Next: how it makes power, one stage at a time.**

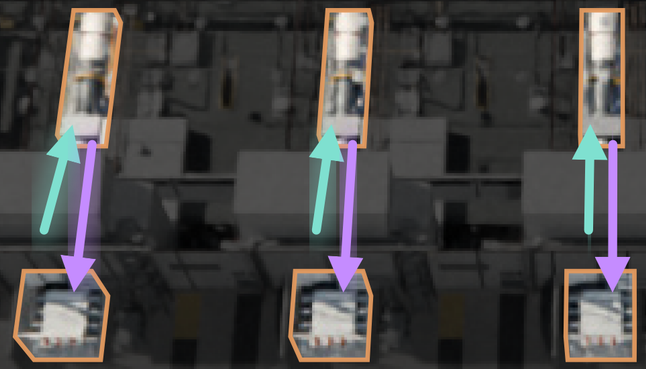


stephan.hardt@southwire.com 

AIR POWER

# THE GAS TURBINE: A JET ENGINE ON THE GROUND

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



It burns gas in compressed air; the hot gas spins the turbine and its generator. That makes **about two-thirds** of the plant's power, and leaves exhaust at about 1,100 °F.

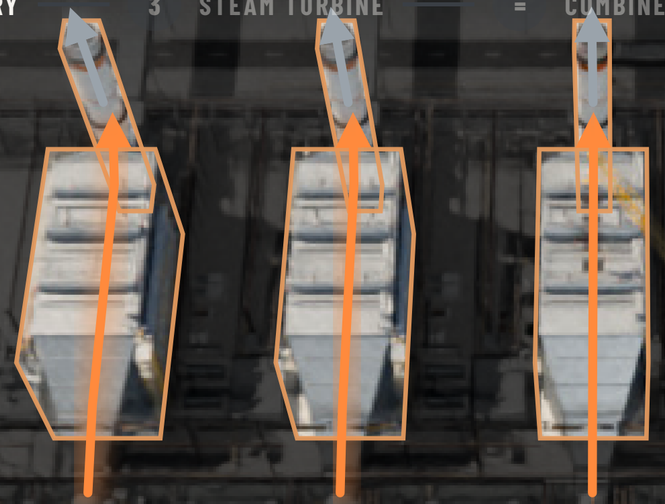


stephan.hardt@southwire.com 

HOT EXHAUST — TO THE STACK

# THE HRSG: CATCH THE HEAT

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



Instead of going up the stack, the hot exhaust runs through a heat-recovery steam generator and boils water into steam. **No extra fuel.**

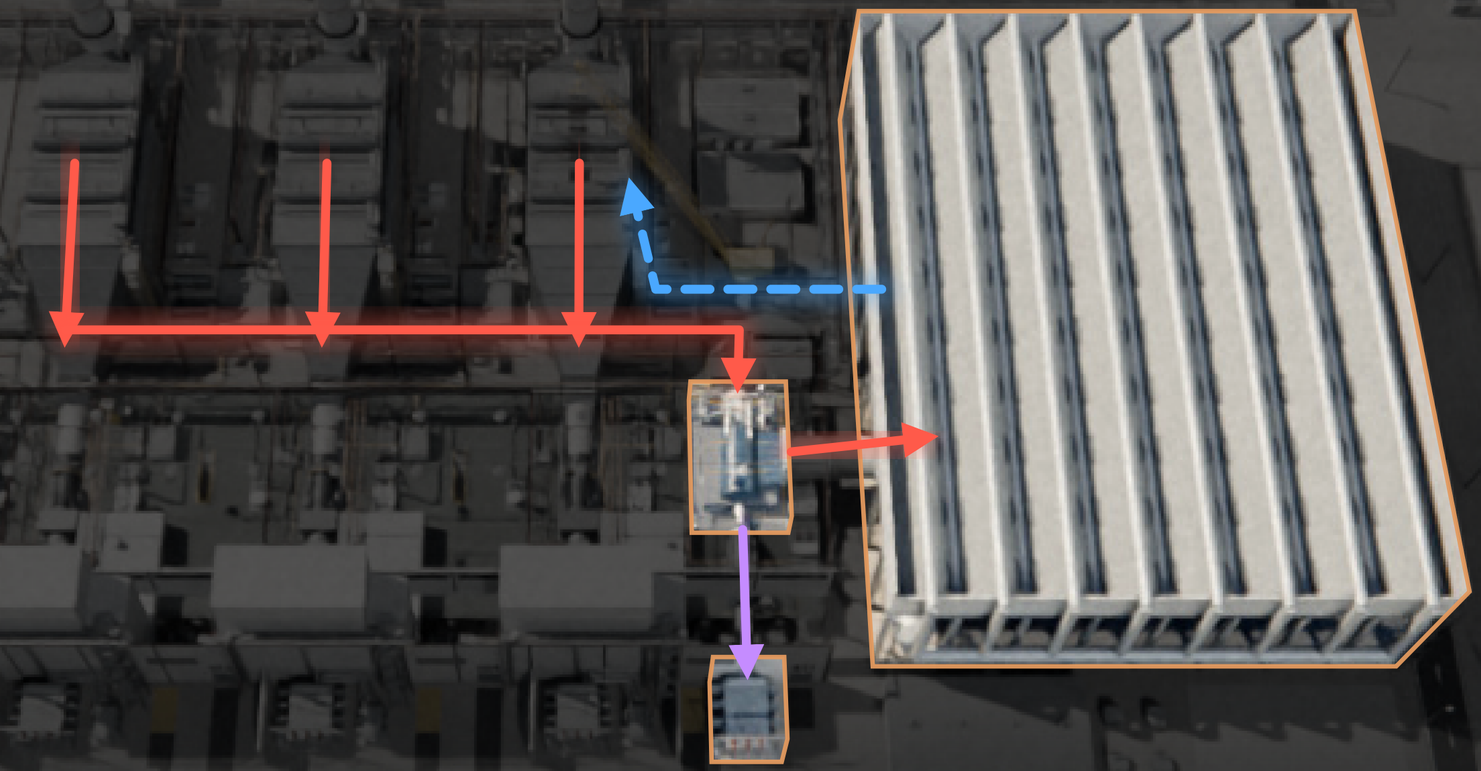


stephan.hardt@southwire.com 

STEAM WATER POWER

# THE STEAM TURBINE: POWER FROM THE LEFTOVERS

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE = COMBINED CYCLE



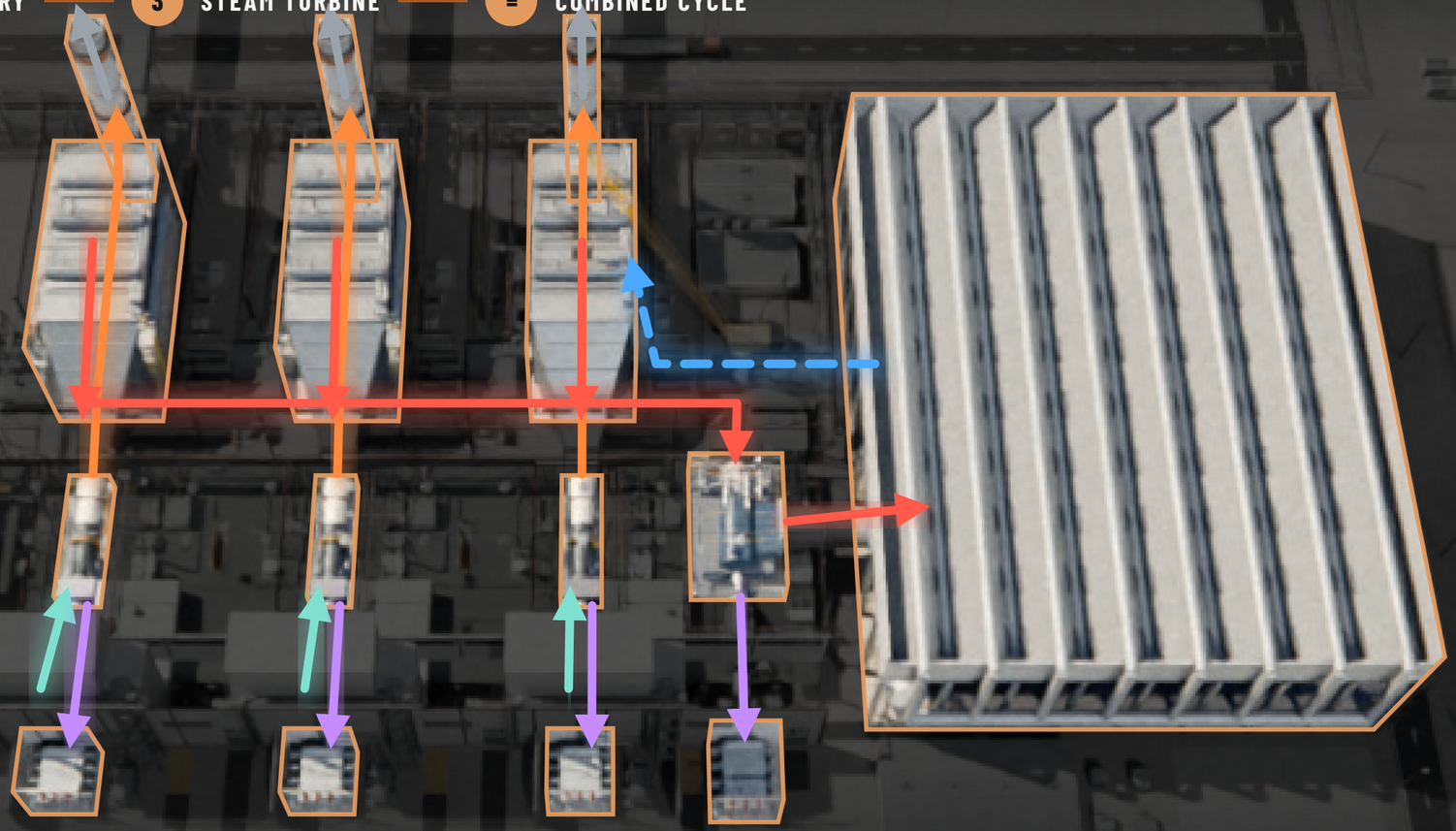
Steam from all three HRSGs spins one more turbine and generator: **about one-third more power.** The air-cooled condenser turns it back into water, and round it goes.

COMBINED CYCLE

# BURN THE GAS ONCE, USE THE HEAT TWICE

EXHAUST · STEAM · WATER · POWER

1 GAS TURBINE — 2 HEAT RECOVERY — 3 STEAM TURBINE — = COMBINED CYCLE



**~60%**
of the fuel becomes power, vs ~40% for a gas turbine alone

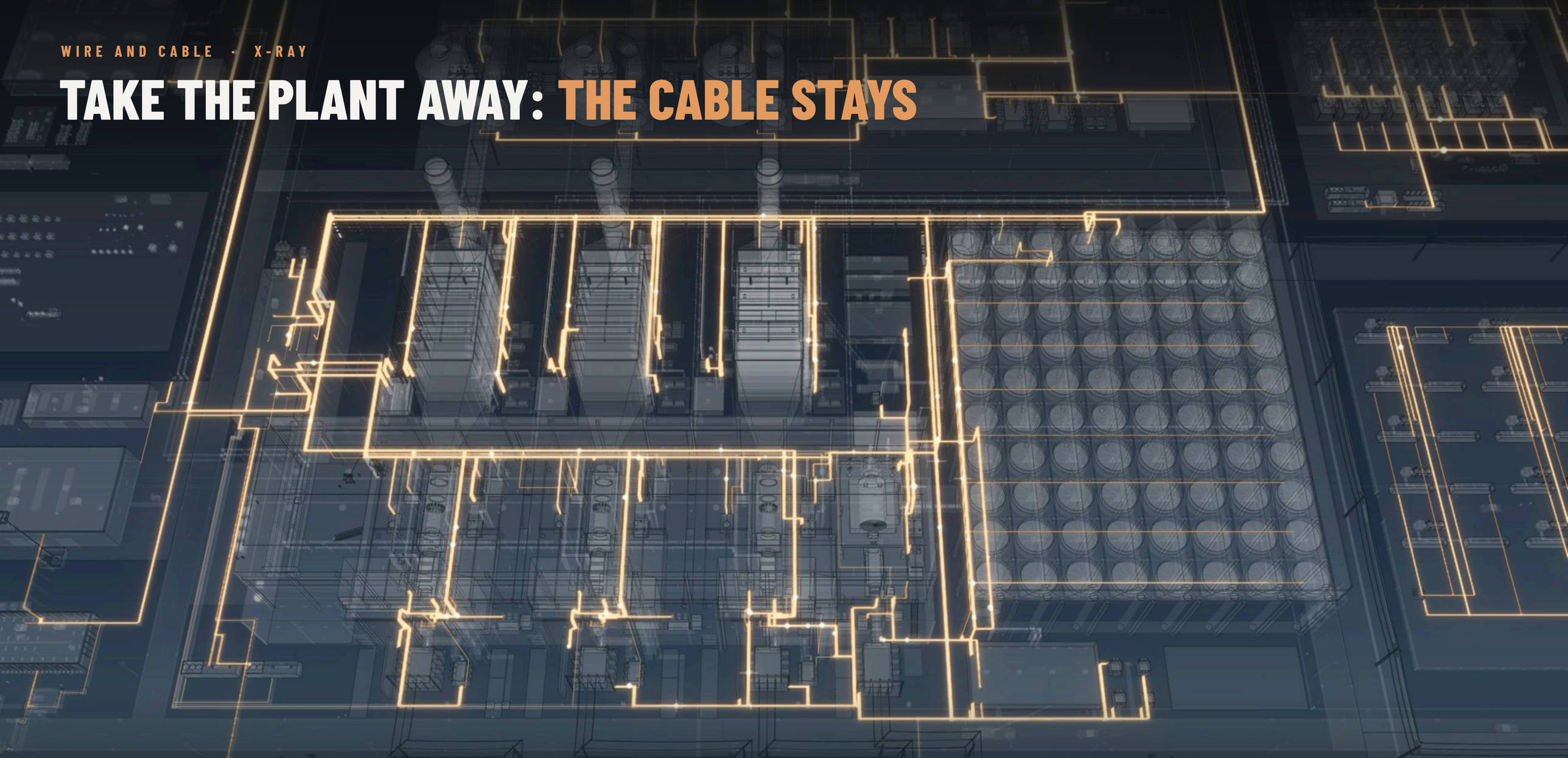
**3 + 1**
three gas turbines, three HRSGs, one steam turbine: a "3x1"

**8,234**
cables in this power block and its condenser, in our model



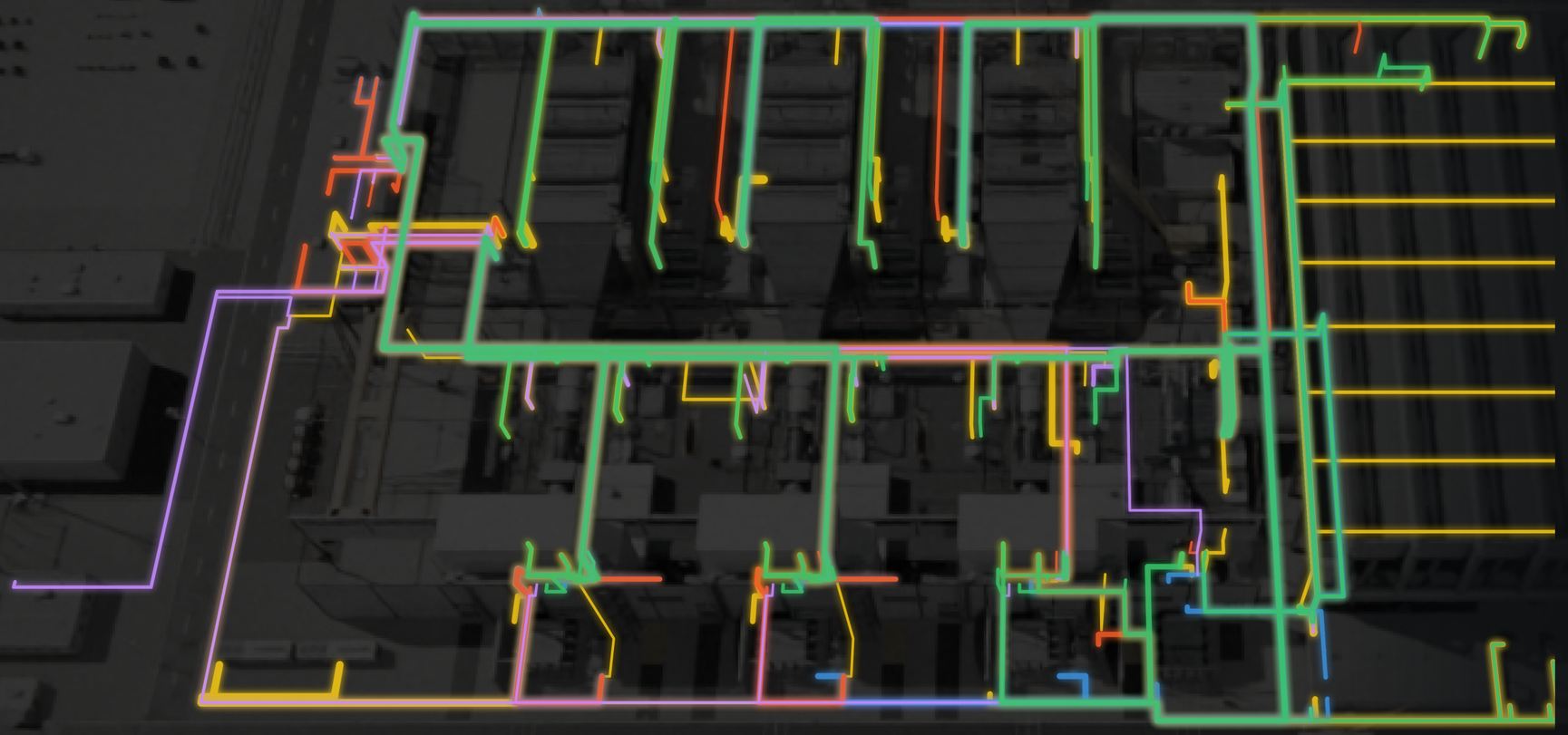
stephan.hardt@southwire.com 

# TAKE THE PLANT AWAY: THE CABLE STAYS



The same view, with the steel made glass. **About 8,200 cables and 440 miles** in the power block and its condenser, every one routed in our model.

# NOW SORT IT BY JOB: FIVE KINDS OF CABLE



Every line is a route from the model's cable schedule, in trays, conduit and duct banks; thicker lines carry more cables. **The power block is a web of copper.**

## THE CABLE IN THIS PICTURE

**8,234** cables · ~439 miles

**MV / HV POWER** — 232 · 28 mi
The big feeders: generators, transformers, the largest motors. MV-105, HV XLPE.

**LV POWER** — 1,160 · 95 mi
600 V to motors, valves, drives, lighting. TC-ER, ARMOR-X MC-HL, VFD cable.

**CONTROL** — 1,255 · 94 mi
Starts, stops and trips: breakers, interlocks, MCC starters. Multi-conductor TC-ER.

**INSTRUMENTATION** — 5,484 · 202 mi
The plant's nerves: pressure, flow, temperature. Shielded pairs, thermocouple extension.

**FIBRE AND FIRE ALARM** — 103 · 21 mi
The control network, protection and life safety.

The power block and its air-cooled condenser only, from the SW-3X1 model cable schedule (building wire, grounding and overhead not counted). Conceptual estimate, not engineered.



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# FROM THE GENERATOR TO THE LIGHT SWITCH

Power is made at medium voltage, raised to high voltage to travel, then stepped down again near the customer.

1

## GENERATOR

**13.8–25 kV**

21 kV in SW-3X1

CABLE

isolated-phase bus, MV-105

2

## STEP-UP (GSU)

**→ 115–765 kV**

230 kV in SW-3X1

CABLE

HV XLPE cable, switchyard bus

3

## TRANSMISSION

**115–765 kV**

long distance, overhead

CABLE

overhead conductor, OPGW

4

## SUBSTATION

**→ 4–35 kV**

steps it down for the town

CABLE

substation cable, grounding

5

## DISTRIBUTION

**4–35 kV**

feeders along the streets

CABLE

MV URD, overhead distribution

6

## CUSTOMER

**480 / 240 / 120 V**

factory, office, home

CABLE

building wire, MC, service entrance

← INSIDE THE PLANT: TODAY

> **Higher voltage, lower current, smaller losses:** that is why transmission runs at hundreds of kV. Southwire has a product family at every step, from the generator to the outlet.

# LEVELS, RATINGS AND THE CABLE THEY DRIVE

One HRSG train from the generator down to the field signals: each voltage level, the equipment and its rating, what it feeds, and the cable that follows.

| LEVEL | EQUIPMENT · RATING | FEEDS · CURRENT | CABLE IT DRIVES |
|---|---|---|---|
| 230 kV | **GSU-1, 21 / 230 kV step-up** | Plant to the grid, overhead line | **Overhead conductor (outside the HRSG scope)** |
| 21 kV | **GT1 generator, isolated-phase bus; UAT-1 21 / 13.8 kV, 40 MVA** | UAT-1 secondary 1,674 A into SWGR-13.8-1 | **4 sets of 3 x 1/C 750 kcmil MV-105** |
| 13.8 kV | **SWGR-13.8-1: metal-clad, 1,200 A, 40 kA, arc-resistant** | Feed pumps BFP-1A / 1B, 5,000 kW each, direct-on-line, 232 A; T-R2A primary, 105 A | **3 x 1/C 250 kcmil MV-105 per pump; 3 x 1/C #2 to T-R2A** |
| 480 V | **T-R2A 2,500 kVA → LC-R2A 4,000 A main, 65 kA → MCC-R2A-1 / -2, 800 A, 65 kA; VFDs** | Secondary 3,007 A; SCR blowers 2 x 110 kW (156 A); 45 HRSG MOVs; feed-pump and chemical-feed auxiliaries; welding receptacles | **10 sets of 4 x 1/C 500 kcmil; 3/0 TC-ER / MC-HL VFD; #10 TC-ER to each MOV** |
| 120 V AC / 125 V DC | **UPS-R2A 10 kVA; 125 V DC battery, 60 cells; DC and UPS panels** | DCS, F&G node, breaker trip and close, burner-management power | **1/C #6 and control TC-ER 14 AWG** |
| Signals | **4-20 mA / HART, thermocouple millivolts, 24 V DC discretes** | Drum level, pressures, flame scanners, gas-path and tube-metal temperatures | **439 instrument + 84 thermocouple-extension cables, shielded pairs** |

**750 kW**
connected at 480 V on one HRSG train, 78 circuits

**2 x 5 MW**
feed pumps: the largest motors on the train, at 13.8 kV

**3,007 A**
on the T-R2A secondary: 40 cables of 500 kcmil

**74 %**
of the train's cables are instrument or thermocouple pairs (68 % of the footage)

**8.4 / 24.6**
cal/cm$^2$ arc flash at 13.8 kV / 480 V (equipment labels)

Ratings are the model's typical values (Load List and nameplates), not engineered. Real plants vary: many put the feed pumps on a 4.16 or 6.9 kV bus.



stephan.hardt@southwire.com 

# THE CABLE FAMILIES IN A POWER PLANT

From the biggest to the smallest. The whole SW-3X1 campus: 21,217 cables, 1,212 miles.

### HV — 69–345 kV

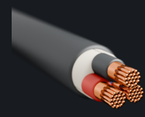
XLPE: switchyard and gen-tie

### MV — 5–35 kV

MV-105: big motors, transformers, feeders

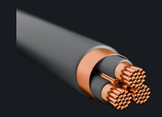
### LV POWER — 600 V

TC-ER, MC, ARMOR-X: motors, panels, lighting



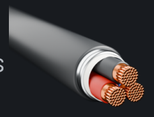
### VFD — 600 V–2 kV

drive-rated: fans and pumps on drives



### CONTROL — 600 V

TC-ER multi-conductor: breakers, valves, interlocks



### INSTRUMENTATION — 300 V

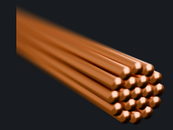
PLTC / ITC shielded pairs, thermocouple extension

### FIBRE — –

controls network, protection

### GROUNDING — bare Cu

the ground grid under everything



**By length:** Instrumentation 44% · LV 600 V power 20% · Control 18% · Fibre, fire alarm, other 11% · MV / HV 7%. The big cables get the attention; the small ones make up most of the miles.

Shares by footage from the SW-3X1 model cable schedule; voltage ratings typical.

# WHAT A POWER PLANT DOES TO A CABLE

| STRESS | WHERE | THE ANSWER |
|---|---|---|
| **HEAT** | beside turbines, HRSGs and steam lines | high-temperature ratings, route away from hot surfaces |
| **CYCLING** | daily starts: heat, vibration, moving terminations | flexible strands, good terminations, support |
| **ELECTRICAL NOISE** | variable-frequency drives on fans and pumps | VFD-rated, shielded cable |
| **HAZARDOUS AREAS** | gas skids and fuel yards (Class I, Div 2) | ARMOR-X MC-HL, no conduit |
| **FIRE AND SMOKE** | halls, control rooms, cable spreading rooms | flame-retardant, low-smoke, circuit integrity |
| **OUTDOORS** | sun, oil, water, burial | sunlight- and oil-resistant jackets, direct-burial ratings |
| **SCHEDULE** | long pulls on the critical path | SIMpull reels, cut-to-length, pre-terminated |

**The wrong cable fails early, and a cable failure can trip a unit.** That is why the spec is strict, and why it is written for each area of the plant.



stephan.hardt@southwire.com 

# SAME PHYSICS, DIFFERENT OWNERS

The owner decides how the project is structured, and how it gets bought. An independent power producer (IPP) builds each plant as its own company, sells the power on contract or into a market, and funds it with investors.

## THE STRUCTURAL EQUATION

**THERMAL IPP =** **PROJECTCO** (one company per plant) + **ASSET** (conversion plant) + **CONTRACT** (power purchase agreement) + **CUSTOMER** (off-taker / load) + **CAPITAL** (investors, tax equity) + **MARKET** (ERCOT, PJM, MISO)

## POWER OWNER ARCHETYPES

### IOUS / CO-OPS — REGULATED

- Utility-led, rate-based ownership
- Own and operate generation
- Buy like utilities: standards, cycles

### HYBRID IPPS — MIXED MODEL

- Utility-affiliated IPP subsidiaries
- Bridge regulated and competitive
- Two buying behaviours

### PURE IPPS — COMPETITIVE

- Privately owned, investor-backed projects
- A direct fit with the project equation
- Speed and cost drive decisions

**OUR SALES FOCUS FOR COMPETITIVE THERMAL POWER**

# GREENFIELD AND BROWNFIELD: FIVE KINDS OF SITE

## GREENFIELD · NEW SITES 137 GW in the register

### 77.0 GW

### GREENFIELD, GRID-CONNECTED

A new plant on open land, tied to the grid.

Everything is new: the full cable schedule, the switchyard and an HV gen-tie to the POI.

### 54.5 GW

### DATA-CENTRE / BEHIND-THE-METER CAMPUS

Power built next to the load it serves.

Modular engines, turbines and batteries; MV collection to the campus; fast schedules, bought by package builders.

## BROWNFIELD · EXISTING SITES 69 GW in the register

### 41.3 GW

### COAL-SITE REPOWER

New gas units on a retired coal or boiler site.

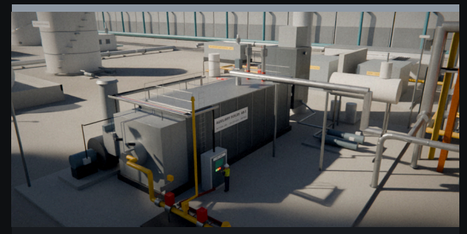
Reuses the land, water and grid connection; new plant cable tied into an old switchyard.

### 21.4 GW

### EXPANSION AT AN OPERATING PLANT

New units beside ones that keep running.

Work around live equipment; tie-ins wait for outages; shared switchyard and controls.



### 11.6 GW

### INDUSTRIAL CAPTIVE AND SITE REUSE

Power, often with steam, for a factory or refinery.

Hazardous areas (MC-HL), tight routes, the host plant's own standards.

**And on plants already running:** outages, HRSG module replacements, upgrades and conversions. There the outage clock is the cost, so cable is staged, pre-terminated and pulled fast. Utilities mostly reuse old sites; IPPs and behind-the-meter developers mostly build new.

Gross GW by site class, Southwire PowerGen register, Sep 11, 2026 (205.8 GW eligible, COD 2026–2035). Site classes provisional (Rev01).

# ONE COMPANY SPECIFIES THE CABLE, ANOTHER ONE BUYS IT

## DEVELOP

site, permits, interconnection request, fuel and water

**WHO**
Owner / developer

**YEARS: THE INTERCONNECTION QUEUE IS OFTEN THE LONG POLE**

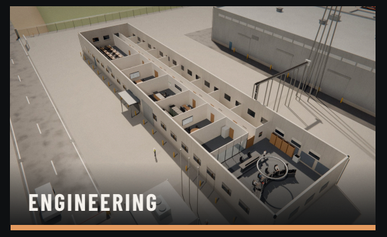
## FINANCE

PPA or rate case, turbine slot, financing decision (FID)

**WHO**
Owner, lenders, OEM

**TURBINES ARE RESERVED YEARS AHEAD**

## ENGINEER




owner's engineer writes the specs; the EPC does detailed design

**WHO**
Owner's engineer, EPC

**CABLE IS SPECIFIED HERE**

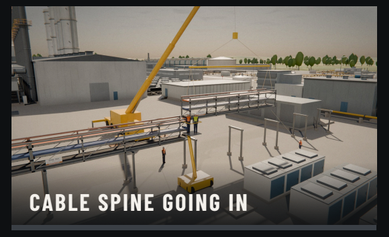
## PROCURE

EPC and package builders buy equipment and bulk materials

**WHO**
EPC, OEMs, package builders

**CABLE IS BOUGHT HERE**

## BUILD




civil, mechanical, then electrical: trays, pulls, terminations

**WHO**
EPC and electrical contractor

**CABLE PULLS SIT ON THE CRITICAL PATH**

## OPERATE

commercial operation (COD), outages, upgrades, repowering

**WHO**
Owner / operator

**REPLACEMENT AND BROWNFIELD WORK**

**The owner pays, the owner's engineer writes the spec, the EPC or package builder places the order.** Be in the spec early, and be ready when the EPC buys: by then the schedule is tight and cable is on the critical path.

# THREE BUCKETS OF CABLE, THREE DIFFERENT BUYERS

## 1 INSIDE THE PACKAGES

WHO BUYS

**OEMS, E-HOUSE AND PACKAGE BUILDERS**

E-house internals (switchgear and MCC interconnections, control and panel wire, short ties), turbine skids, modular gensets

WHEN

Every month, across many projects, months before site

OUR PRODUCTS

MV-105 ties, TC-ER, control and panel wire, pre-terminated harness kits

**NOT IN OUR 440 MILES: A SEPARATE MARKET ON TOP**

## 2 BETWEEN THE PACKAGES

WHO BUYS

**THE EPC OR ITS ELECTRICAL CONTRACTOR**

Field cable from the e-houses out to every motor, valve, instrument and transformer: the X-ray

WHEN

Once per plant, when detailed design fixes the cable schedule

OUR PRODUCTS

MV-105, TC-ER, ARMOR-X MC-HL, VFD, PLTC / ITC, thermocouple; SIMpull, cut-to-length

**MOST OF THE MILES: ~440 IN THE POWER BLOCK**

## 3 TO THE GRID

WHO BUYS

**THE EPC'S HV CONTRACTOR OR THE UTILITY**

GSU to switchyard, the gen-tie to the POI, grounding

WHEN

Early: HV cable has the longest lead time

OUR PRODUCTS

HV XLPE (69-345 kV), bare copper grounding

**FEW CABLES, HIGH VALUE**

**Starting in the e-house is not the same as being bought by the e-house builder.** Cover both: EPCs for the field cable; package and e-house builders for the cable inside. The builders buy every month and shape the spec at the terminations.

# WHERE EVERY CABLE STARTS

The R1 switchgear room from above, roof off. **Power leaves the switchgear and MCCs; signals come home to the control room.** Every device in between is wired.

## SWITCHGEAR

13.8 and 4.16 kV lineups: every MV feeder starts here

## MCCS AND DRIVES

480 V starters and VFDs: motors, fans and pumps

## CONTROL ROOM

where the instrument and control cable comes home

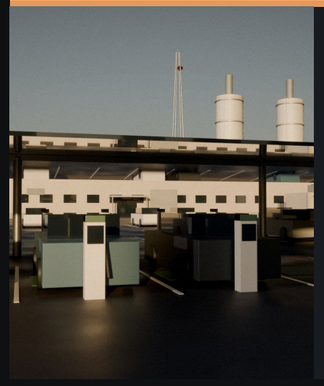
# A MODERN GAS PLANT IS MORE THAN THE POWER BLOCK

## PRIVATE 5G / LTE

A monopole with massive-MIMO antennas: the plant's own wireless network.

**CABLE**

fibre backhaul, tower power, grounding



## EV CHARGING + SOLAR CARPORT

Chargers at every bay, a DC fast charger, PV on the carport roof.

**CABLE**

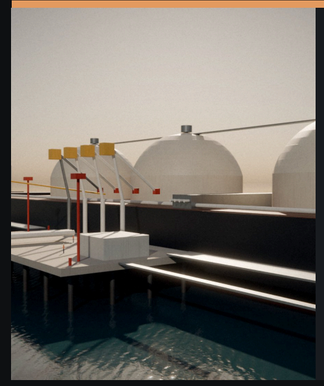
LV feeders, PV wire, a pad-mount transformer

## BEHIND-THE-METER DATA CENTRE

A ~150 MW campus fed straight from the plant and the modular yard.

**CABLE**

34.5 kV collection, cable bus, UPS feeders



## LNG BY SEA

A marine terminal: the carrier at the berth, unloading arms, the tank onshore.

**CABLE**

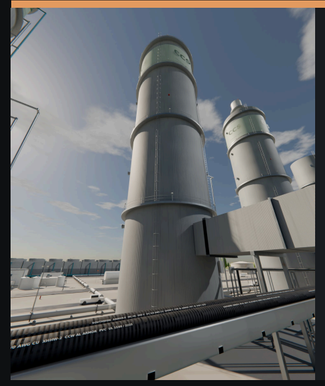
hazardous-area cable, long runs to the jetty, MV pumps

## HYDROGEN

Tube-trailer bay and receiving station; an electrolyzer building beside it.

**CABLE**

Class I Div 2 runs, gas detection, DC cable



## CARBON CAPTURE

Absorber, stripper and compressors on the stack exhaust.

**CABLE**

a second plant's motors, drives and instruments

**All six are in the SW-3X1 model.** Each one adds its own network of cable on top of the power block: the campus is the customer, not just the turbines.

# 04

# THE TOUR

Seven stops on the SW-3X1 campus: what it does, its electrical system, its cable.

THE CAMPUS · FROM THE SEA TO THE GRID

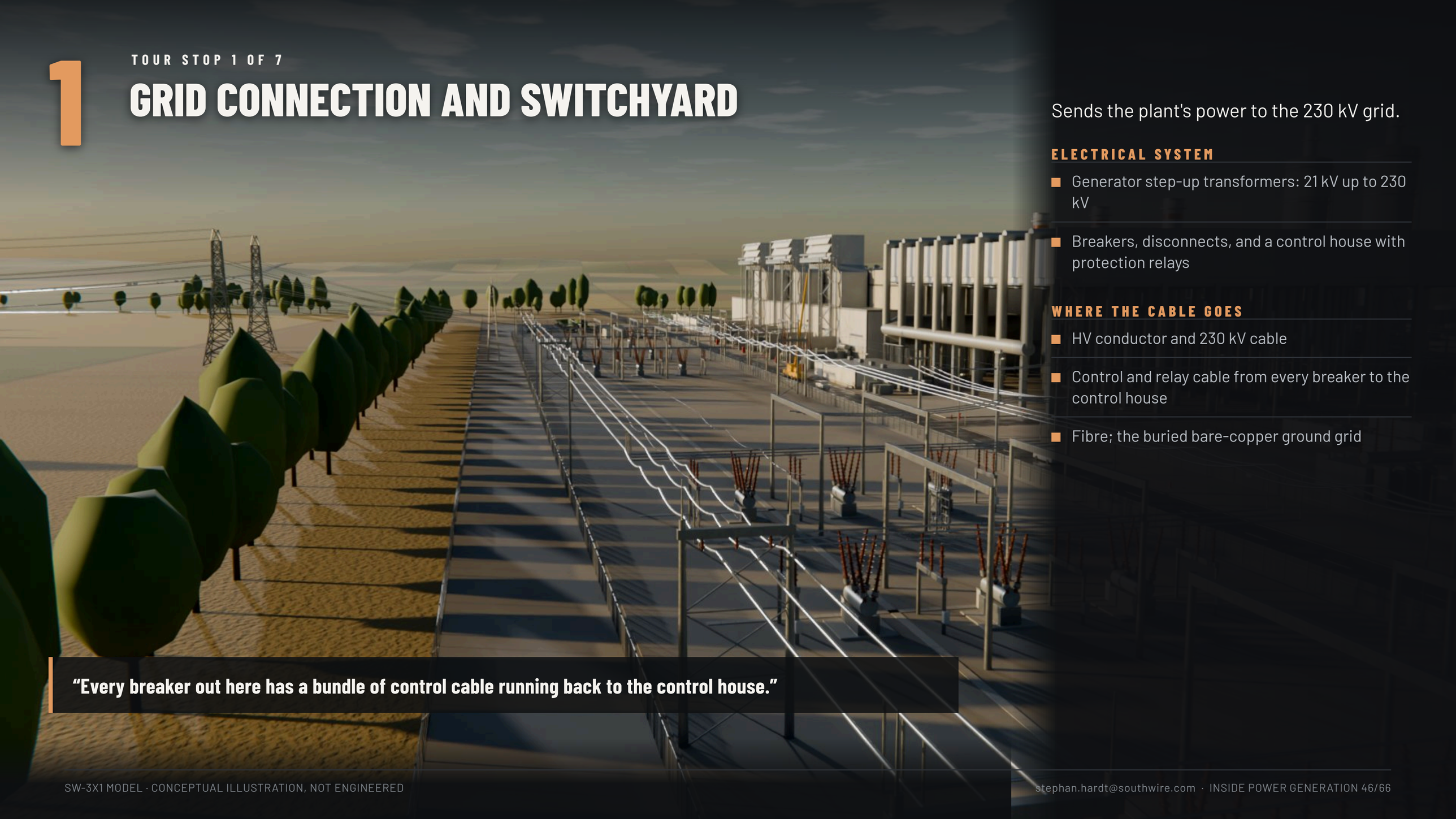
# ONE SITE, SEVEN STOPS

1. GRID CONNECTION AND SWITCHYARD
2. TURBINE HALL
3. HRSG AND ITS E-HOUSE
4. COOLING AND WATER
5. MODULAR GENERATION
6. BEHIND THE METER: THE DATA CENTRE
7. FUEL: GAS YARD AND LNG

LNG arrives by ship at the marine terminal; the power leaves on 230 kV lines. Everything in between runs on cable.

TOUR STOP 1 OF 7

# 1 GRID CONNECTION AND SWITCHYARD



Sends the plant's power to the 230 kV grid.

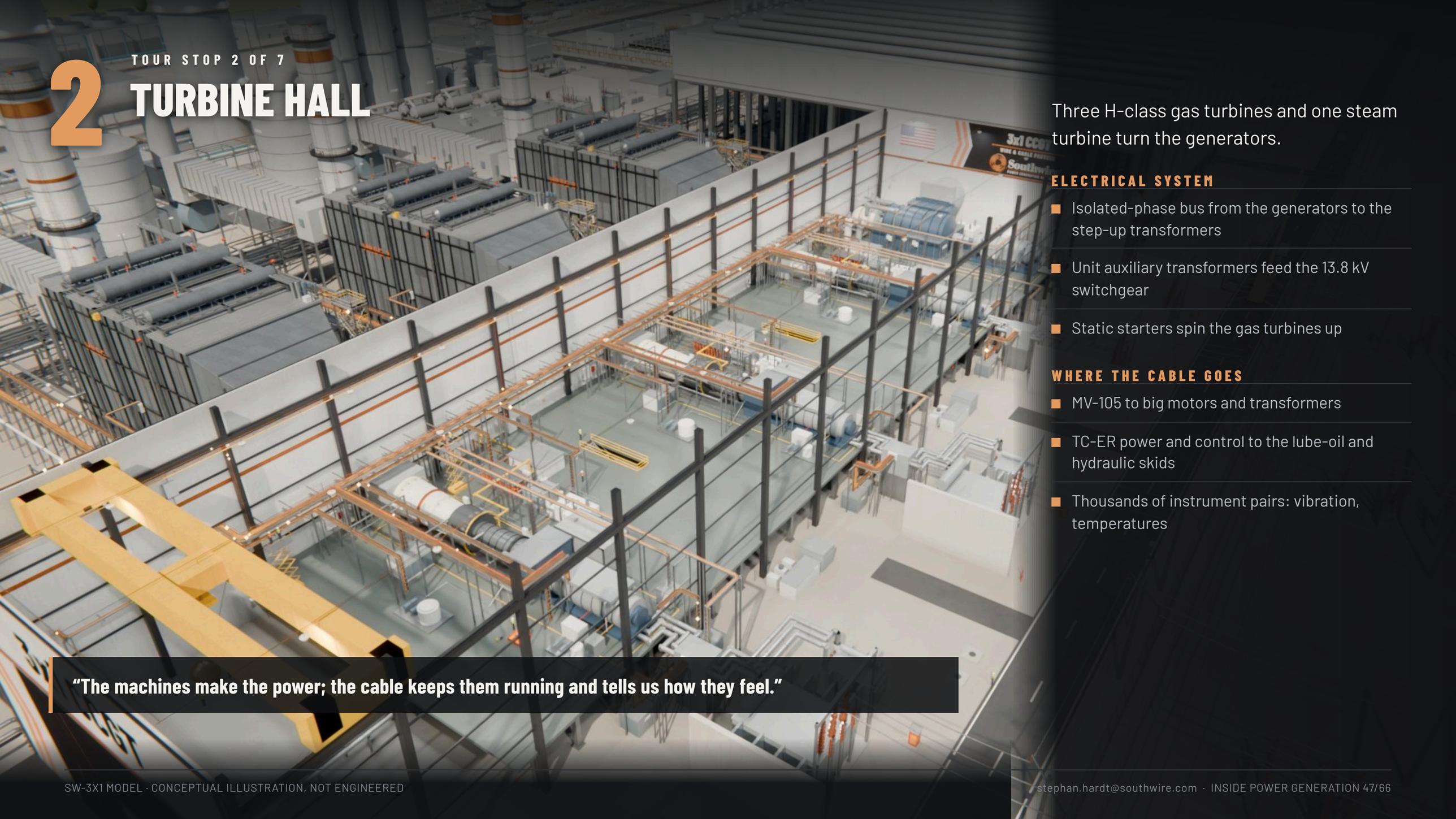
## ELECTRICAL SYSTEM

- Generator step-up transformers: 21 kV up to 230 kV
- Breakers, disconnects, and a control house with protection relays

## WHERE THE CABLE GOES

- HV conductor and 230 kV cable
- Control and relay cable from every breaker to the control house
- Fibre; the buried bare-copper ground grid

**“Every breaker out here has a bundle of control cable running back to the control house.”**



TOUR STOP 2 OF 7

# 2 TURBINE HALL

Three H-class gas turbines and one steam turbine turn the generators.

## ELECTRICAL SYSTEM

- Isolated-phase bus from the generators to the step-up transformers
- Unit auxiliary transformers feed the 13.8 kV switchgear
- Static starters spin the gas turbines up

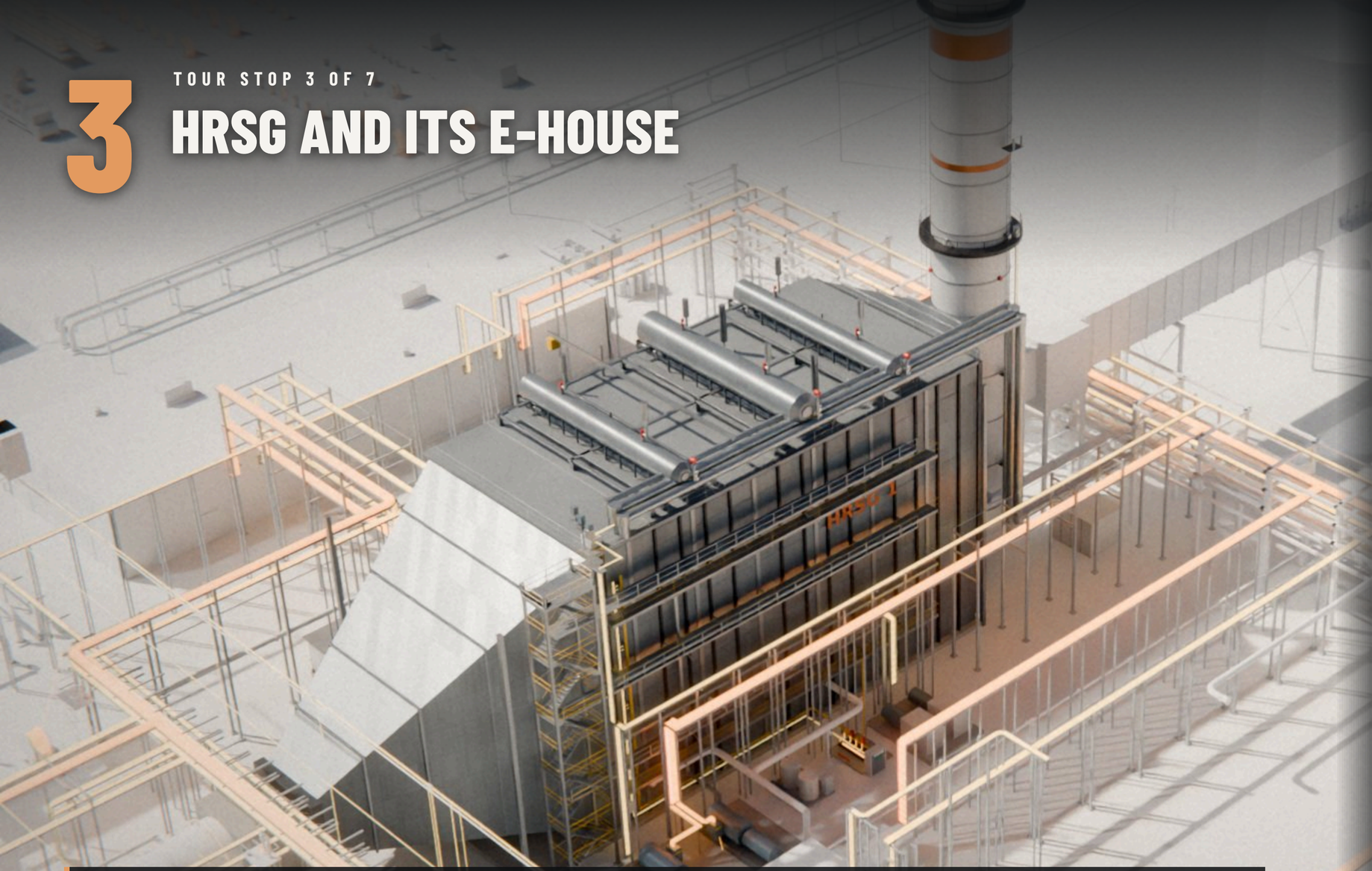
## WHERE THE CABLE GOES

- MV-105 to big motors and transformers
- TC-ER power and control to the lube-oil and hydraulic skids
- Thousands of instrument pairs: vibration, temperatures

**“The machines make the power; the cable keeps them running and tells us how they feel.”**

TOUR STOP 3 OF 7

# 3 HRSG AND ITS E-HOUSE




Turns turbine exhaust heat into steam; the SCR inside cuts NOx.

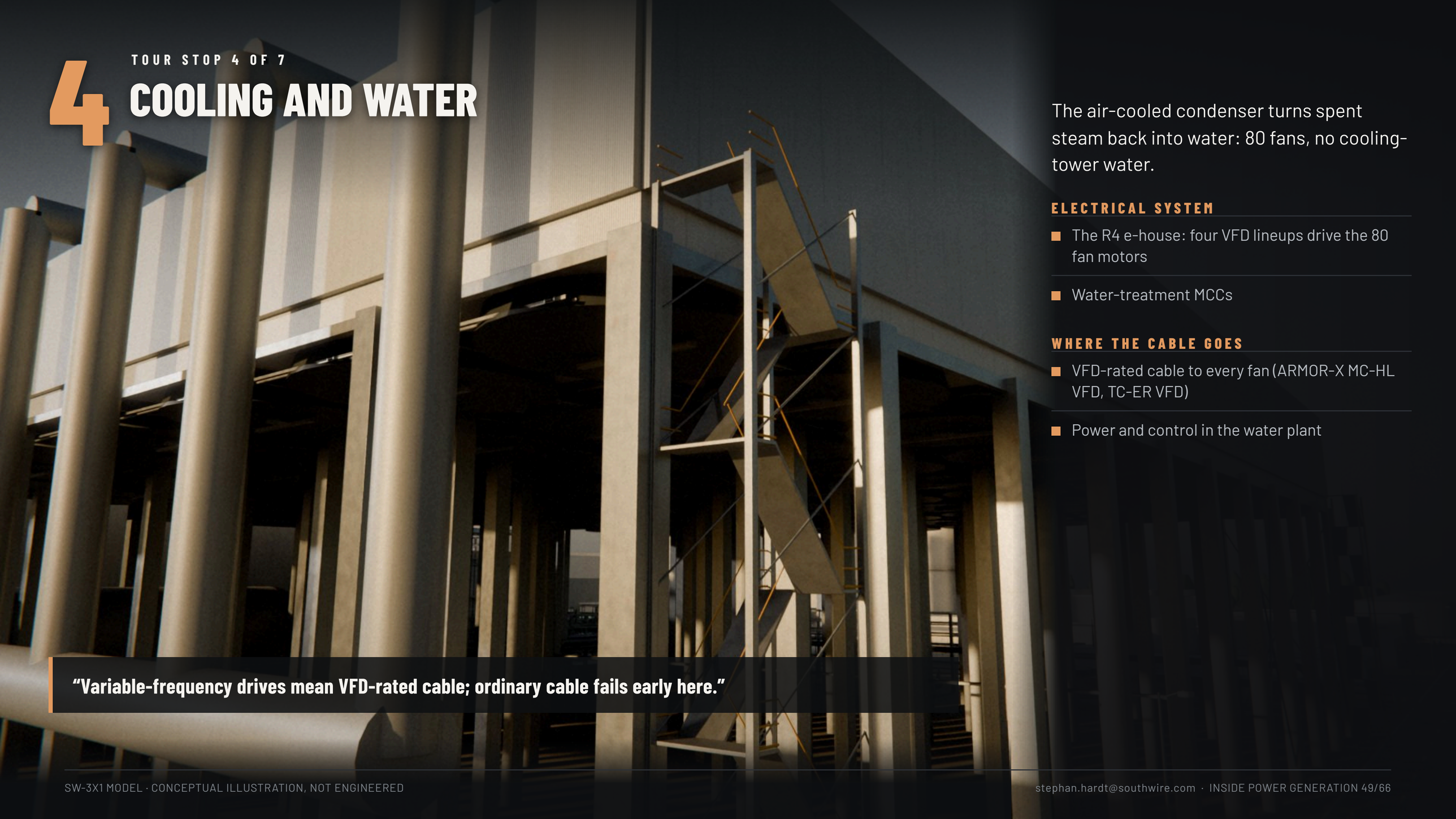
## ELECTRICAL SYSTEM

- Two 13.8 kV feed pumps
- A transformer feeds the e-house: 480 V switchgear, MCCs, drives, DCS
- CEMS shelter at the stack

## WHERE THE CABLE GOES

- One HRSG train: 707 cables, about 157,000 ft
- About three quarters are instrument and thermocouple pairs
- ARMOR-X MC-HL at the duct-burner gas skid

**“One boiler train is about 30 miles of cable, and most of it is small instrument cable.”**



TOUR STOP 4 OF 7

# 4 COOLING AND WATER

The air-cooled condenser turns spent steam back into water: 80 fans, no cooling-tower water.

## ELECTRICAL SYSTEM

- The R4 e-house: four VFD lineups drive the 80 fan motors
- Water-treatment MCCs

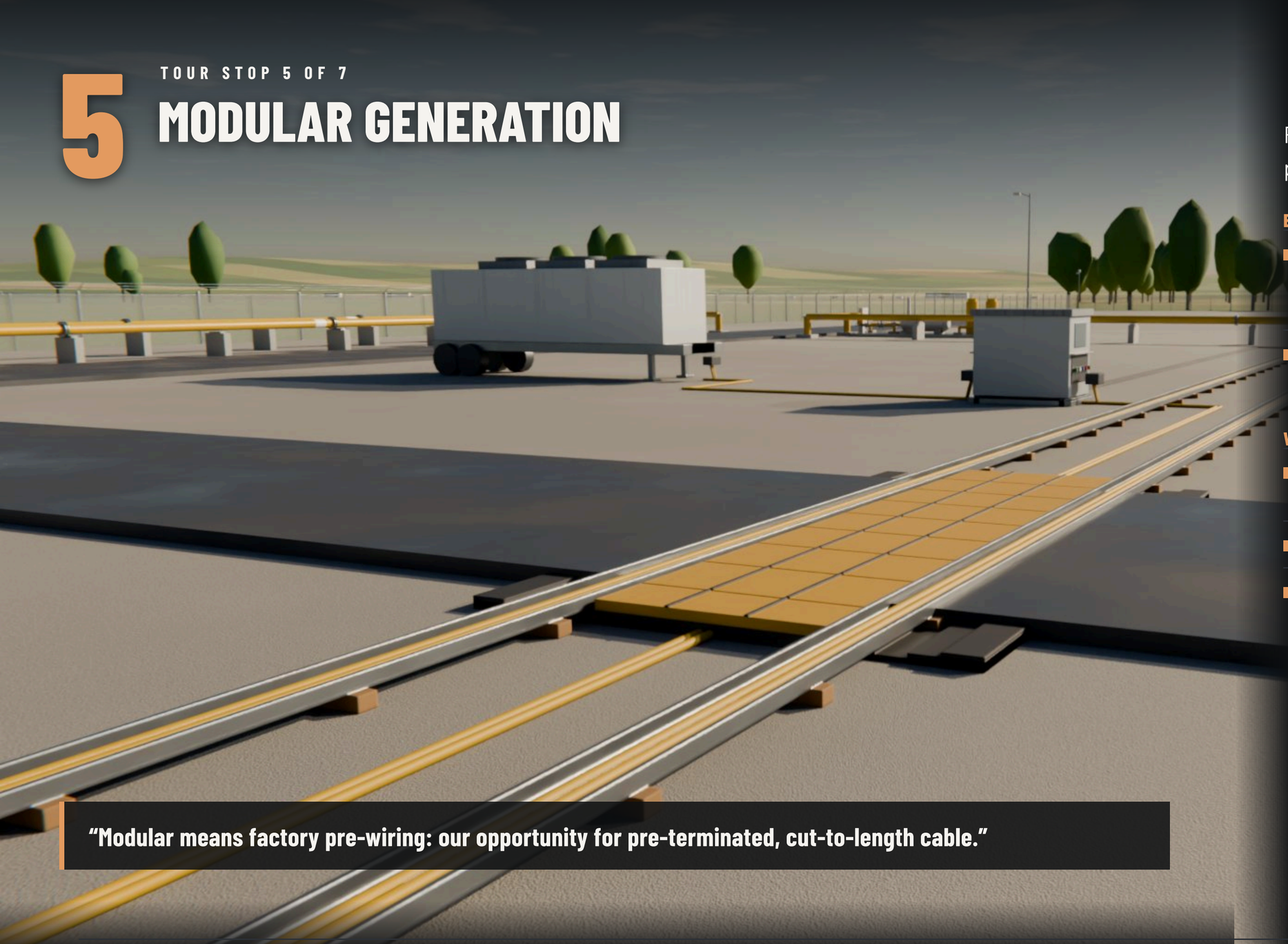
## WHERE THE CABLE GOES

- VFD-rated cable to every fan (ARMOR-X MC-HL VFD, TC-ER VFD)
- Power and control in the water plant

**“Variable-frequency drives mean VFD-rated cable; ordinary cable fails early here.”**

TOUR STOP 5 OF 7

# 5 MODULAR GENERATION



Fast-start, flexible power beside the big plant.

## ELECTRICAL SYSTEM

- Gas engines in a hall, aeroderivative and industrial gas-turbine packages, containers, trailers, fuel cells, microturbines
- Each unit on the 13.8 kV collector; inverters for the battery and fuel cells

## WHERE THE CABLE GOES

- MV collection cable; DC cable at the battery and fuel cells
- Factory pre-wired skids and plug-in modules
- Pre-terminated and cut-to-length cable

**“Modular means factory pre-wiring: our opportunity for pre-terminated, cut-to-length cable.”**

TOUR STOP 6 OF 7

# 6 BEHIND THE METER: THE DATA CENTRE

A data-centre campus powered directly by the plant's modular units, with the grid as backup.

## ELECTRICAL SYSTEM

- 13.8 to 34.5 kV step-up transformers and double-bus switchgear
- A 40 MW battery absorbs load steps
- Two data halls fed 2N through pad-mount transformers and UPS

## WHERE THE CABLE GOES

- 35 kV MV cable
- Large LV feeders to the UPS skids
- Fibre everywhere

> **“Data centres are why gas is growing again, and they are cable-dense on both the A and the B side.”**

TOUR STOP 7 OF 7

# 7 FUEL: GAS YARD AND LNG

Brings the fuel in and conditions it; an LNG satellite holds about three hours of backup fuel.

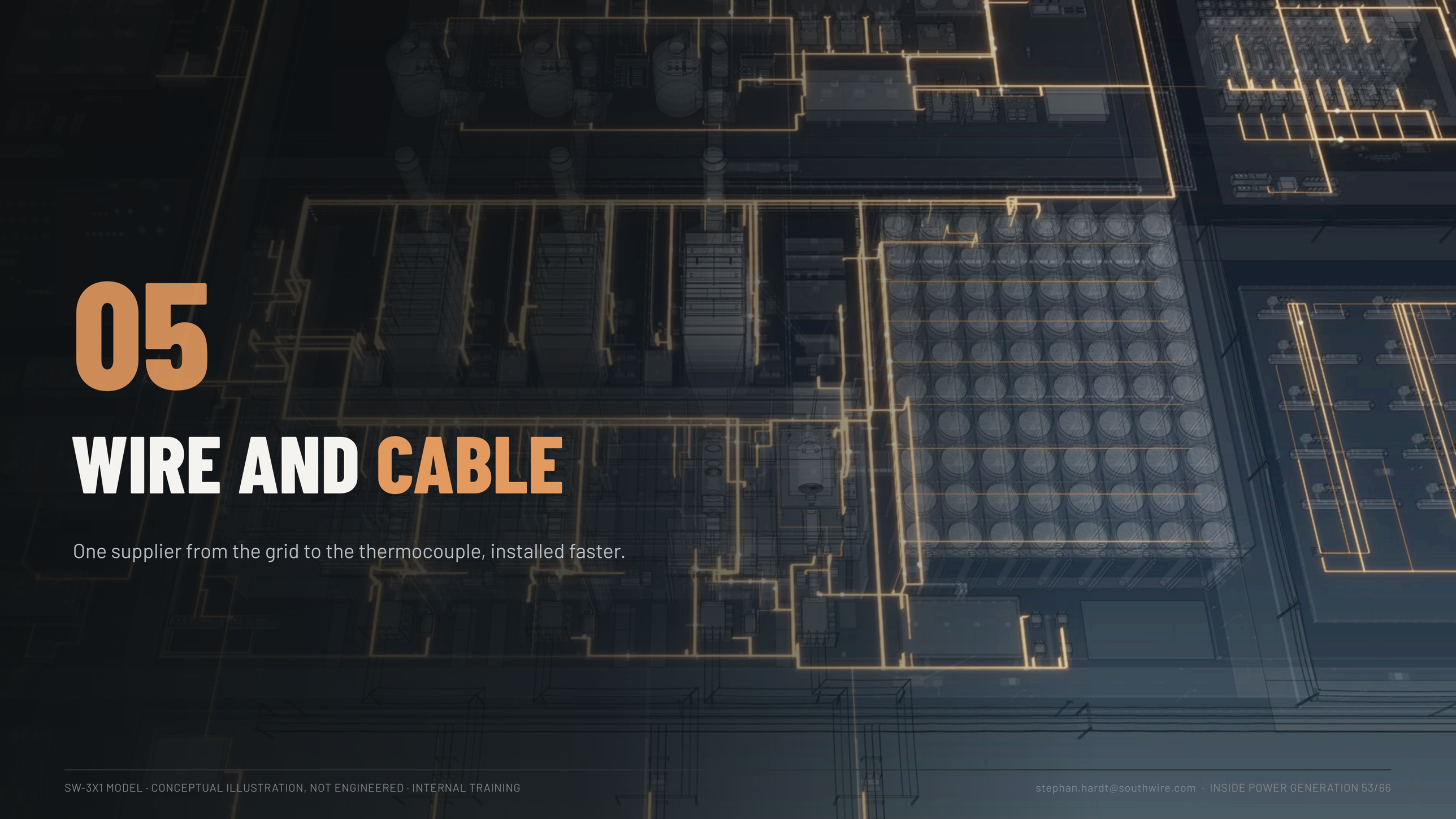
## ELECTRICAL SYSTEM

- Metering and regulating station; the LNG e-house (13.8 kV / 480 V)
- Pumps, vaporizers, boil-off compressor

## WHERE THE CABLE GOES

- ARMOR-X MC-HL for Class I Div 1 and 2
- TC-ER control, PLTC instrumentation and fire alarm on one strut rack

> **“Wherever there's gas, the cable has to be rated for it.”**



# 05

# WIRE AND CABLE

One supplier from the grid to the thermocouple, installed faster.

# THE WHOLE CAMPUS IN ONE NUMBER

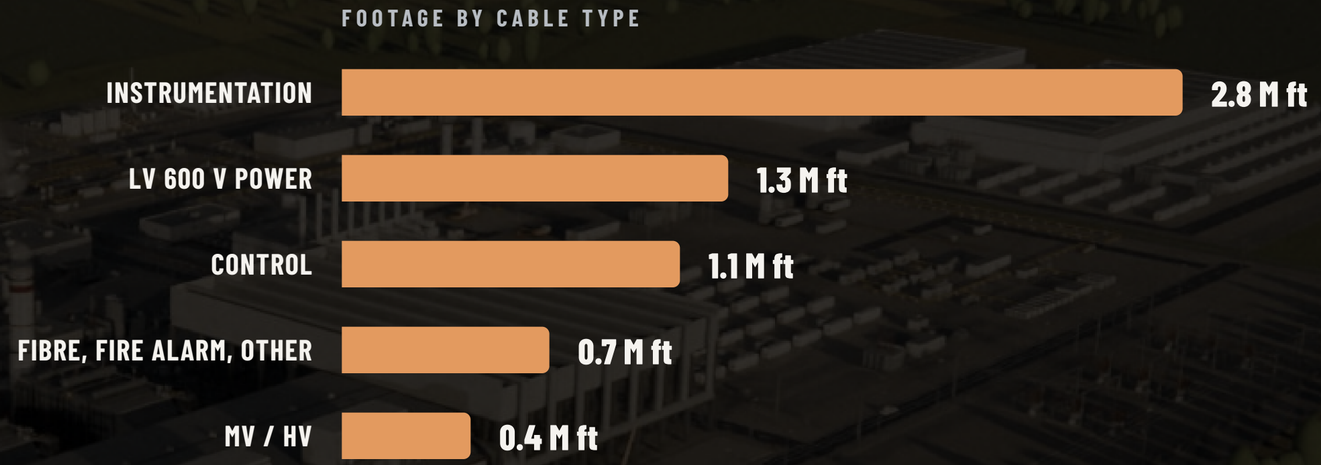
**6.4M**

FEET OF CABLE · ABOUT 1,212 MILES

**6,011**
CABLE RUNS

**21,217**
CABLES

From the model's cable schedule: every circuit on the campus, from the 230 kV switchyard to the last thermocouple.




Most of the footage is small cable: instrument and control. MV is the high-value part, but it isn't the volume.

# ONE SUPPLIER, GRID TO THERMOCOUPLE

| | |
|---|---|
| **GRID CONNECTION** | HV XLPE transmission cable (230 kV) · bare copper grounding |
| **POWER ISLAND: MV** | MV-105 EPR 133% (5–35 kV) · ARMOR-X MC-HL 15 kV |
| **POWER ISLAND: LV AND DRIVES** | Type TC-ER power · ARMOR-X MC-HL 600 V · VFD-rated cable |
| **CONTROL AND INSTRUMENTATION** | TC-ER control · PLTC / ITC shielded pairs · thermocouple extension |
| **HAZARDOUS AREAS** | ARMOR-X MC-HL, Class I Div 2: gas skids, fuel yards, burner fronts |
| **BATTERY STORAGE** | PV / RHW-2 DC conductor (2 kV) · MV-105 collection (35 kV) |
| **MODULAR AND TEMPORARY POWER** | Type SHD-GC portable MV · Type W · cam-lock sets |
| **INSTALLATION** | SIMpull reels and SIMpull truck · cut-to-length · pre-terminated harness kits |

**Plus the services:** Spec support: sizing, derating, spec reviews · Cut-to-length against the cable schedule · Harness kits, pre-terminated and labelled · Project logistics: reel tracking, sequenced deliveries · Field support: pulling calcs, terminations, tests.

Typical Southwire families for each application; confirm constructions, sizes and listings against current Southwire specifications.



stephan.hardt@southwire.com 

WIRE AND CABLE · INSTALLED FASTER

# ACCELERATING TIME TO POWER

Cable is decided in the field, on the critical path. Every reel set-up and every termination saved is schedule saved.

### SIMPULL REELS

Several conductors paralleled on one reel: one pull per feeder instead of one per conductor.

### SIMPULL TRUCK

Delivery, payoff and reel return in one vehicle.

### CUT-TO-LENGTH

Reels cut and labelled to the cable schedule, pull by pull: less waste, no field measuring.

### PRE-TERMINATED

Factory-made, tested harnesses for skids and panels: on the critical path you test instead of terminate.

THE REEL YARD: SIMPULL REELS STAGED FOR THE PULLS

KNOWLEDGE CHECK · ZOOM POLL

# THREE QUESTIONS, ONE MINUTE

## 1 WHICH CABLE TYPE HAS THE MOST MILES IN THE POWER BLOCK?

- A MV / HV POWER
- B LV POWER
- C INSTRUMENTATION
- D FIBRE

## 2 WHO USUALLY PLACES THE CABLE ORDER?

- A THE OWNER
- B THE OWNER'S ENGINEER
- C THE EPC OR PACKAGE BUILDER
- D THE UTILITY

## 3 IN A COMBINED CYCLE, THE STEAM TURBINE ADDS ABOUT HOW MUCH MORE POWER, WITH NO EXTRA FUEL?

- A 5%
- B ONE-THIRD
- C DOUBLE
- D NONE

THREE THINGS TO REMEMBER

# TAKE AWAY

**1** **GAS IS GROWING AGAIN**

driven by data centres and the need for firm power.

**2** **MOST OF A PLANT'S CABLE IS SMALL**

instrument and control; the big MV and power cable is the smaller share.

**3** **SOUTHWIRE COVERS THE WHOLE PLANT**

one supplier, staged on SIMpull reels, spec-compliant from MV down to the thermocouple.

Southwire®
POWER GENERATION SOLUTIONS

# Q&A

Questions from the chat. The recording and the 3D viewer link will be shared after the session.

**Stephan Hardt**
stephan.hardt@southwire.com

AFTER TODAY

# RESOURCES AND WHO TO CALL

## RESOURCES

### THE 3D PLANT MODEL

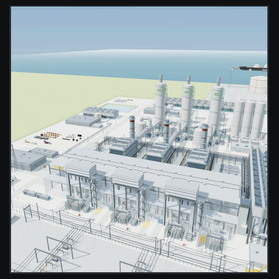
The interactive viewer you saw today, with the Nov 11 tour and the cable X-ray. Link sent with the recording.

### THE BROCHURES

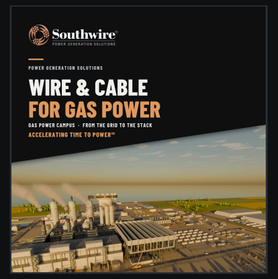
Wire & cable for gas power · EPC solutions · boiler solutions.

### THE RECORDING AND THESE SLIDES

Shared after the session, with a glossary of the terms used today.

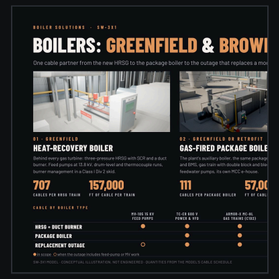


3D VIEWER




WIRE & CABLE BROCHURE

EPC SOLUTIONS




BOILER SOLUTIONS

## POWER GENERATION SOLUTIONS TEAM

| | |
|---|---|
| **STEPHAN HARDT** | PowerGen strategy, the plant model, owners and OEMs |
| **PATRICK RITCHIE** | EPC accounts |
| **GREYSON** | EPC outreach to more gas-plant builders |
| **PHIL METZ · BOB BENISH** | OEMs and package builders: e-houses, switchgear, modular power |

stephan.hardt@southwire.com

# B

# BACKUP SLIDES

Extra detail if questions go there, and the sources.

# THE ANSWERS AND WHY

## 1 WHICH CABLE TYPE HAS THE MOST MILES IN THE POWER BLOCK?

A MV / HV POWER | B LV POWER | **C INSTRUMENTATION** | D FIBRE

Instrumentation: about 200 of the 440 miles. The big cables get the attention; the small ones are most of the miles.

## 2 WHO USUALLY PLACES THE CABLE ORDER?

A THE OWNER | B THE OWNER'S ENGINEER | **C THE EPC OR PACKAGE BUILDER** | D THE UTILITY

The EPC or package builder buys it; the owner's engineer specifies it. Be in the spec early, be ready when the EPC buys.

## 3 IN A COMBINED CYCLE, THE STEAM TURBINE ADDS ABOUT HOW MUCH MORE POWER, WITH NO EXTRA FUEL?

A 5% | **B ONE-THIRD** | C DOUBLE | D NONE

About one-third: the HRSGs turn the gas turbines' hot exhaust into steam. Burn the gas once, use the heat twice.

# FOUR WAYS TO BURN GAS FOR POWER

| | SIMPLE CYCLE<br>THE PEAKER | COMBINED CYCLE<br>THE WORKHORSE | ENGINES (RICE)<br>THE SPRINTER | AERODERIVATIVE<br>THE JET |
|---|---|---|---|---|
| UNIT SIZE | **50–400 MW per gas turbine** | **300–1,500+ MW per block** | **2–20 MW per engine, in banks** | **30–60 MW per unit** |
| EFFICIENCY | **~35–40%** | **~55–64%** | **~40–50%** | **~38–42%** |
| START TO FULL LOAD | **10–30 min** | **about 1–3 hours** | **about 5 min** | **5–10 min** |
| RUNS AS | Peaking, mid-merit | Baseload, mid-merit | Peaking, data-centre firming | Peaking, fast backup |
| IN THE MODEL | A gas turbine alone: exhaust goes straight up the stack | Gas turbines + HRSGs + steam turbine: the SW-3X1 power block | Big gas engines in a hall: the modular yard | A jet engine on the ground: the modular yard |

**Units that stick:** 1 GW = 1,000 MW = 1,000,000 kW. **MW** is power (how hard it runs right now); **MWh** is energy (power × hours). Efficiency is how much of the fuel's energy leaves as electricity; the rest goes up the stack as heat.

Typical ranges for new US units; efficiency on a lower-heating-value basis. Starts vary with design and how hot the unit is.



stephan.hardt@southwire.com 

BACKUP · FUNDAMENTALS · THE BASICS

# HOW AN HRSG WORKS

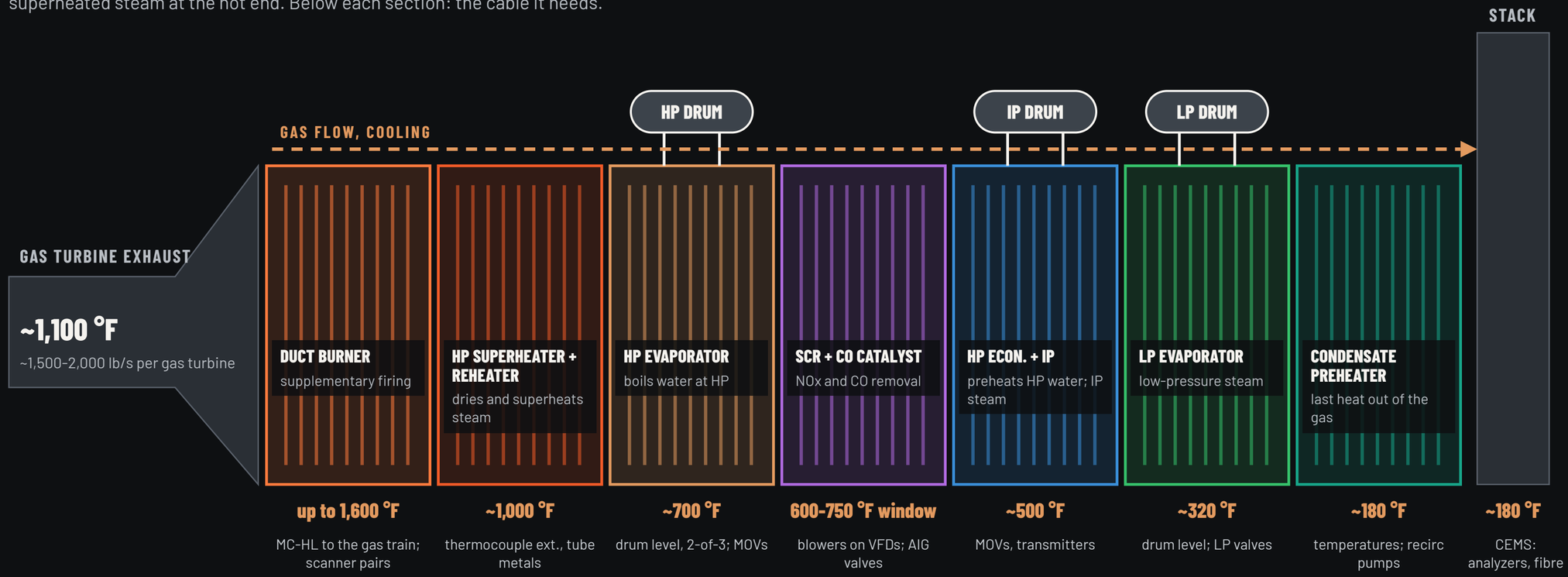
The exhaust crosses rows of finned tubes, hottest sections first; water enters at the cold end and leaves as superheated steam at the hot end. Below each section: the cable it needs.

## WHAT IT IS

- **Without it:** ~60 % of the fuel's energy goes up the stack as ~1,100 °F exhaust (~40 % efficient).
- **With it:** a boiler heated by that exhaust; its steam drives the steam turbine.
- **The payoff:** ~1/3 more power from the same fuel (~60 %+ efficient).

## FUN FACTS

- Fuel, CO2 and cost per MWh drop by about a third.
- The NOx catalyst (SCR) lives **inside** it, where the gas is just the right temperature.
- Duct burners can fire extra gas for peak demand.
- No HRSG: a gas-turbine plant. With one: combined cycle.




Steam to the steam turbine, typical: HP ~1,800-2,400 psig, ~1,050 °F · reheat / IP ~400-600 psig, ~1,050 °F · LP ~50-80 psig

**~1/3**
of a combined-cycle plant's output comes from HRSG steam, with no extra fuel (duct firing aside)

**~60 %+**
combined-cycle efficiency, against ~40 % for the gas turbine alone

**3 x 3**
three pressures, three drums; each drum level measured three ways, typically voted 2-out-of-3

**~1 hour**
a hot start: about an hour: every start cycles the tubes, which is why tube-metal thermocouples matter

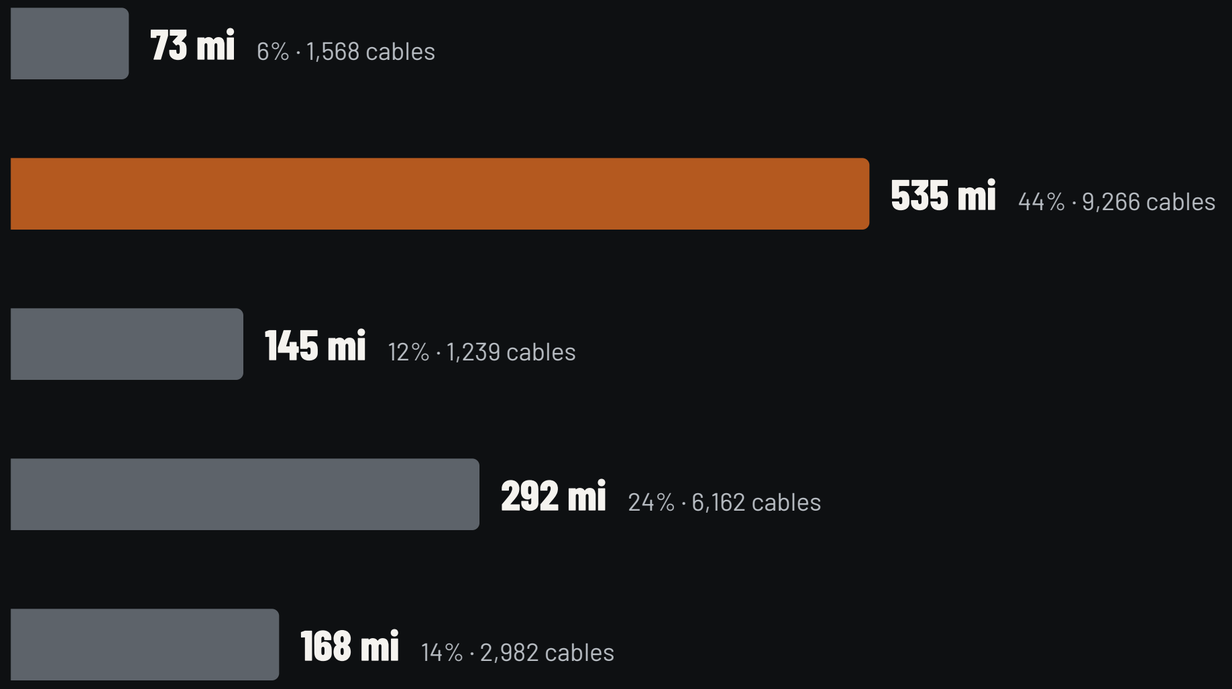
Typical three-pressure reheat HRSG with SCR behind an H-class gas turbine; section order, temperatures and pressures vary by design.



stephan.hardt@southwire.com · INSIDE POWER GENERATION 63/66

# FOLLOW THE POWER: THE CABLE IS EVERYWHERE

THE WHOLE SW-3X1 CAMPUS BY PLANT AREA, FROM THE MODEL'S CABLE SCHEDULE (1,212 MILES)




### THE POI

**Point of interconnection:** where the plant meets the grid. The plant side is the owner's and EPC's scope; past it, the utility or transmission owner builds and buys.

The gen-tie (often HV underground) runs from the switchyard to the POI, and the scope line there is often blurred: a place we help.

**Most of the cable is inside the box**: thousands of instrument, control and motor circuits in the turbine hall and HRSGs. The grid connection carries the biggest cables; the power block carries the most.

# WHAT THE SPEC CALLS FOR

Every cable application on the HRSG, the requirements an owner's or EPC's cable specification writes for it, and the Southwire product built to meet them.

| APPLICATION · WHERE | WHAT THE SPECIFICATION CALLS FOR | SOUTHWIRE |
|---|---|---|
| **MV POWER, 15 KV**<br>FEED-PUMP MOTORS; THE T-R2A PRIMARY | 1/C Cu, EPR insulation, **133 % level**, 90 °C, tape shield, PVC jacket. ICEA S-93-639, UL 1072 (MV-105). Terminations IEEE 48; acceptance test VLF (IEEE 400.2) / NETA ATS | **MV-105 15 kV, 220 mil NL-EPR 133 %, SIMpull · SPEC 46302** |
| **480 V FEEDERS AND POWER**<br>T-R2A SECONDARY, LC-R2A TO MCCS, MOTORS AND MOVS | Cu XHHW-2, 90 °C wet / dry, sized at **125 %** of full load; voltage drop ≤ 3 %. ICEA S-95-658, UL 44 / UL 1277; 10+ parallel sets: compare with bus duct | **XHHW-2 1/C (SIMpull) · Type TC-ER 3/C + ground · SPEC 45252** |
| **TRAY CABLE: POWER AND CONTROL**<br>EVERY TRAY BETWEEN THE E-HOUSE AND THE HRSG | Type TC-ER (UL 1277) so it may leave the tray as an exposed run to the equipment (NEC 336.10); sunlight resistant, direct burial; **IEEE 1202 / UL 1685** 70,000 Btu/h vertical-tray flame test | **Control TC-ER 14 AWG multiconductor · SPEC 45053** |
| **HAZARDOUS AREA: GAS TRAIN, DUCT BURNER**<br>DUCT-BURNER FUEL SKID, BMS PANEL TO THE VALVES | Class I Div 2 wiring methods per NEC 501.10(B); MC-HL (UL 1569 + UL 2225) also qualifies for Div 1. Burner management per **NFPA 85**, duct burner UL 295, listed fittings | **ARMOR-X MC-HL, welded armor · SPEC 45352** |
| **VARIABLE-FREQUENCY DRIVES**<br>SCR DILUTION-AIR BLOWERS, FD FANS | VFD-rated: symmetrical three-ground construction, overall shield or continuous armor for the common-mode current, XLPE insulation for reflected-wave peaks | **ARMOR-X MC-HL VFD · TC-ER VFD · SPEC 45352** |
| **INSTRUMENTATION, 4-20 MA / HART**<br>DRUM LEVEL, PRESSURES, FLAME SCANNERS, ANALYZERS | **Individually shielded** twisted pairs + overall shield, drain wires; 300 V PLTC (UL 13, NEC 725) or ITC; ICEA S-73-532; segregated from power in tray | **Instrumentation PLTC, SPOS · SPEC 43303** |
| **THERMOCOUPLE EXTENSION**<br>GAS-PATH, TUBE-METAL AND DUCT TEMPERATURES | Alloy matched to the thermocouple (type K = KX), ANSI MC96.1 / ASTM E230 colour code; **never copper**; shielded pairs; PLTC; 105 °C near the casing | **Thermocouple extension, PLTC multi-pair · SPEC 42345 / 42350** |
| **FIRE ALARM, FIBRE, BUILDING WIRE**<br>F&G LOOPS, CEMS DATA, E-HOUSE LIGHTING | FPLR / FPLP (NEC 760, NFPA 72); single-mode fibre, armored in tray; THHN / THWN-2 in conduit (UL 83) | **SIMpull THHN / THWN-2 · fire-alarm and fibre cable** |

## CHECKED AGAINST A REAL PLANT

A recent US utility combined-cycle filing shows the same structure: **HRSG MCCs A / B** fed from 480 V switchgear through 2,500 / 3,300 kVA transformers off the auxiliary bus (6.9 kV there, 13.8 kV here).

## DESIGN AND INSTALL

Raceway and cable design for generating stations: IEEE 422. Grounding: IEEE 80 / 665. Area classification: NEC 500, API RP 500. Field acceptance: NETA ATS.



stephan.hardt@southwire.com 

APPENDIX

# SOURCES AND QUALIFIERS

| | |
|---|---|
| **ELECTRICITY MIX 2025; PLANNED 2026 ADDITIONS** | EIA, Electricity explained (2025 utility-scale generation); EIA Today in Energy, Feb 2026 |
| **LOAD GROWTH, DATA CENTRES** | Grid Strategies, National Load Growth Report, 2025 (forecast) |
| **378 GW PIPELINE, 52 GW BUILDING** | Global Energy Monitor, Aug 2026; early-stage proposals will not all be built |
| **TURBINE LEAD TIMES** | S&P Global Commodity Insights, May 2025 |
| **OEM BACKLOGS AND MODULAR ORDERS** | OEM releases, Q2-Q3 2026; Utility Dive, Aug 2026; POWER, Oct 2025 and Feb 2026 |
| **CAPACITY ADDED PER YEAR, 1996-2026** | EIA and Southwire provisional series, to be verified |
| **CABLE OPPORTUNITY, REGISTER, STATES, SITE CLASSES** | Southwire PowerGen planning model and register, Sep 11, 2026: assumed rates and probabilities |
| **PLANT TYPES, GRID VOLTAGES, UNITS** | Typical industry ranges; vary by design |
| **CABLE COUNTS AND MILES** | SW-3X1 model cable schedule: conceptual, not engineered |
| **PLANT AND YARD RENDERS** | SW-3X1 plant model: conceptual and illustrative, not engineering data |

**VERIFY ALL FIGURES AGAINST PRIMARY SOURCES BEFORE EXTERNAL USE**



stephan.hardt@southwire.com 